

| TOTAL EQUIPMENT | TOTAL ACTIVE CHANNELS |
|-----------------|-----------------------|
| 42 Units        | 85 Channels           |

| Item # | Equipment Tag             | Equipment Service Description                                            | P&ID Drawing No. |
|--------|---------------------------|--------------------------------------------------------------------------|------------------|
| 1      | <a href="#">FSL-40201</a> | Slurry low flow                                                          | 325-01-402-PD-01 |
| 2      | <a href="#">FSL-40202</a> | Slurry low flow                                                          | 325-01-402-PD-01 |
| 3      | <a href="#">FT-40201</a>  | Slurry flowrate & density check transfer line                            | 325-01-402-PD-02 |
| 4      | <a href="#">FT-40202</a>  | Feed Slurry Flow Transmitter for Pipe (PIPE)                             | 325-01-402-PD-02 |
| 5      | <a href="#">HV-40201</a>  | Slurry Drain line                                                        | 325-01-402-PD-01 |
| 6      | <a href="#">HV-40202</a>  | Slurry Drain line                                                        | 325-01-402-PD-01 |
| 7      | <a href="#">HV-40203</a>  | Slurry Drain tank TS-40201                                               | 325-01-402-PD-01 |
| 8      | <a href="#">HV-40204</a>  | Slurry Drain tank TS-40202                                               | 325-01-402-PD-01 |
| 9      | <a href="#">HV-40205</a>  | Slurry Drain line                                                        | 325-01-402-PD-01 |
| 10     | <a href="#">HV-40206</a>  | Slurry Drain tank TS-40203                                               | 325-01-402-PD-01 |
| 11     | <a href="#">LT-40201</a>  | Level transmitter in Spray dryer infeed tank                             | 325-01-402-PD-02 |
| 12     | <a href="#">LT-40202</a>  | Level transmitter in pH Adjustment tank 2                                | 325-01-402-PD-01 |
| 13     | <a href="#">LT-40203</a>  | Level transmitter in Spray dryer infeed tank                             | 325-01-402-PD-02 |
| 14     | <a href="#">PT-40201</a>  | Pressure Transmitter for Inlet Filter Strainer (FSTR-11)                 | 325-01-402-PD-02 |
| 15     | <a href="#">PT-40202</a>  | Feed Slurry Pressure Transmitter for Pipe (PIPE)                         | 325-01-402-PD-02 |
| 16     | <a href="#">PT-60202</a>  | Feed Slurry Pressure Transmitter for High Pressure Pump Outlet (PUPD-11) | 325-01-602-PD-01 |
| 17     | <a href="#">SV-40201</a>  | Water service pH analyzer in pH Adjustment tank 1                        | 325-01-402-PD-01 |
| 18     | <a href="#">SV-40202</a>  | Water service pH analyzer in pH Adjustment tank 2                        | 325-01-402-PD-01 |
| 19     | <a href="#">SV-40203</a>  | Air service pH analyzer in pH Adjustment tank 1                          | 325-01-402-PD-01 |
| 20     | <a href="#">SV-40204</a>  | Air service pH analyzer in pH Adjustment tank 2                          | 325-01-402-PD-01 |
| 21     | <a href="#">XV-40201A</a> | Slurry feed-in pH Adjustment tank 1                                      | 325-01-402-PD-01 |
| 22     | <a href="#">XV-40201B</a> | Slurry feed-out Spray dryer infeed tank                                  | 325-01-402-PD-02 |
| 23     | <a href="#">XV-40201C</a> | Water washing line outlet Spray dryer infeed tank                        | 325-01-402-PD-02 |
| 24     | <a href="#">XV-40201D</a> | CIP water washing Spray dryer infeed tank                                | 325-01-402-PD-02 |
| 25     | <a href="#">XV-40201E</a> | CIP high pressure water washing Spray dryer infeed tank                  | 325-01-402-PD-02 |
| 26     | <a href="#">XV-40201F</a> | Slurry feed-to spray dry tank                                            | 325-01-402-PD-01 |
| 27     | <a href="#">XV-40201G</a> | CIP water washing pH Adjustment tank 1                                   | 325-01-402-PD-01 |
| 28     | <a href="#">XV-40201H</a> | CIP High pressure water washing pH Adjustment tank 1                     | 325-01-402-PD-01 |

|    |                                  |                                                              |                  |
|----|----------------------------------|--------------------------------------------------------------|------------------|
| 29 | <a href="#"><u>XV-40202A</u></a> | Slurry feed-in pH Adjustment tank 2                          | 325-01-402-PD-01 |
| 30 | <a href="#"><u>XV-40202B</u></a> | Slurry feed-out pH Adjustment tank 2                         | 325-01-402-PD-01 |
| 31 | <a href="#"><u>XV-40202C</u></a> | Water washing line outlet pH Adjustment tank 2               | 325-01-402-PD-01 |
| 32 | <a href="#"><u>XV-40202D</u></a> | Slurry circulation pH Adjustment tank 2                      | 325-01-402-PD-01 |
| 33 | <a href="#"><u>XV-40202E</u></a> | Slurry pH Adjustment tank 2 transfer to pH Adjustment tank 1 | 325-01-402-PD-01 |
| 34 | <a href="#"><u>XV-40202F</u></a> | Slurry feed-to spray dry tank                                | 325-01-402-PD-01 |
| 35 | <a href="#"><u>XV-40202G</u></a> | CIP water washing pH Adjustment tank 2                       | 325-01-402-PD-01 |
| 36 | <a href="#"><u>XV-40202H</u></a> | CIP High pressure water washing pH Adjustment tank 2         | 325-01-402-PD-01 |
| 37 | <a href="#"><u>XV-40203B</u></a> | Slurry feed-out Spray dryer infeed tank                      | 325-01-402-PD-02 |
| 38 | <a href="#"><u>XV-40203C</u></a> | Water washing line outlet Spray dryer infeed tank            | 325-01-402-PD-02 |
| 39 | <a href="#"><u>XV-40203D</u></a> | CIP water washing Spray dryer infeed tank                    | 325-01-402-PD-02 |
| 40 | <a href="#"><u>XV-40203E</u></a> | CIP high pressure water washing Spray dryer infeed tank      | 325-01-402-PD-02 |
| 41 | <a href="#"><u>pHT-40201</u></a> | pH analyzer in pH Adjustment tank 1                          | 325-01-402-PD-01 |
| 42 | <a href="#"><u>pHT-40202</u></a> | pH analyzer in pH Adjustment tank 2                          | 325-01-402-PD-01 |

# ASIN ENGINEERING | JUNCTION BOX LOOP TEST REGISTER: JB-401

Project Sprint 18K TPA Spray Dryer / Jet Cooker | Location: JB-401 | 1 Equipment = 1 Sheet

| DIGITAL INPUT (DI) | DIGITAL OUTPUT (DO) |
|--------------------|---------------------|
| 50 Pts             | 24 Pts              |

| Primary Signal Breakdown          | Total Channels | Primary JB Terminal | PLC Rack / Slot |
|-----------------------------------|----------------|---------------------|-----------------|
| 1x DI                             | 1              | P1-TBDI1-17         | C1/C1S4         |
| 1x DI                             | 1              | P1-TBDI1-18         | C1/C1S4         |
| 2x AI, 1x DI (Flow + Pulse + Pwr) | 3              | P1-TBAI1-1(A/B)     | C1/C1S10        |
| 1x AI, 1x DI (Flow + Pulse + Pwr) | 2              | P1-TBAI1-3(A/B)     | C1/C1S10        |
| 1x DI (Solenoid + FB)             | 1              | P1-TBDI1-21         | C1/C1S4         |
| 1x DI (Solenoid + FB)             | 1              | P1-TBDI1-22         | C1/C1S4         |
| 1x DI (Solenoid + FB)             | 1              | P1-TBDI1-23         | C1/C1S4         |
| 1x DI (Solenoid + FB)             | 1              | P1-TBDI1-24         | C1/C1S4         |
| 1x DI (Solenoid + FB)             | 1              | P1-TBDI1-25         | C1/C1S4         |
| 1x DI (Solenoid + FB)             | 1              | P1-TBDI1-26         | C1/C1S4         |
| 1x AI                             | 1              | P1-TBAI1-5(A/B)     | C1/C1S10        |
| 1x AI                             | 1              | P1-TBAI1-6(A/B)     | C1/C1S10        |
| 1x AI                             | 1              | P1-TBAI1-4(A/B)     | C1/C1S10        |
| 1x AI                             | 1              | P1-TBAI1-9(A/B)     | C1/C1S10        |
| 1x AI                             | 1              | P1-TBAI1-10(A/B)    | C1/C1S10        |
| 1x AI                             | 1              | P1-TBAI1-11(A/B)    | C1/C1S10        |
| 1x DO (Solenoid + FB)             | 1              | P1-TBRL1-25         | C1/C1S8         |
| 1x DO (Solenoid + FB)             | 1              | P1-TBRL1-26         | C1/C1S8         |
| 1x DO (Solenoid + FB)             | 1              | P1-TBRL1-27         | C1/C1S8         |
| 1x DO (Solenoid + FB)             | 1              | P1-TBRL1-28         | C1/C1S8         |
| 2x DI, 1x DO (Solenoid + FB)      | 3              | P1-TBDI1-27         | C1/C1S4         |
| 2x DI, 1x DO (Solenoid + FB)      | 3              | P1-TBDI1-29         | C1/C1S4         |
| 2x DI, 1x DO (Solenoid + FB)      | 3              | P1-TBDI1-31         | C1/C1S4         |
| 2x DI, 1x DO (Solenoid + FB)      | 3              | P1-TBDI2-1          | C1/C1S5         |
| 2x DI, 1x DO (Solenoid + FB)      | 3              | P1-TBDI2-3          | C1/C1S5         |
| 2x DI, 1x DO (Solenoid + FB)      | 3              | P1-TBDI2-5          | C1/C1S5         |
| 2x DI, 1x DO (Solenoid + FB)      | 3              | P1-TBDI2-7          | C1/C1S5         |
| 2x DI, 1x DO (Solenoid + FB)      | 3              | P1-TBDI2-9          | C1/C1S5         |

|                              |   |                 |          |
|------------------------------|---|-----------------|----------|
| 2x DI, 1x DO (Solenoid + FB) | 3 | P1-TBDI2-19     | C1/C1S5  |
| 2x DI, 1x DO (Solenoid + FB) | 3 | P1-TBDI2-12     | C1/C1S5  |
| 2x DI, 1x DO (Solenoid + FB) | 3 | P1-TBDI2-14     | C1/C1S5  |
| 2x DI, 1x DO (Solenoid + FB) | 3 | P1-TBDI2-16     | C1/C1S5  |
| 2x DI, 1x DO (Solenoid + FB) | 3 | P1-TBDI2-18     | C1/C1S5  |
| 2x DI, 1x DO (Solenoid + FB) | 3 | P1-TBDI2-21     | C1/C1S5  |
| 2x DI, 1x DO (Solenoid + FB) | 3 | P1-TBDI2-23     | C1/C1S5  |
| 2x DI, 1x DO (Solenoid + FB) | 3 | P1-TBDI2-25     | C1/C1S5  |
| 2x DI, 1x DO (Solenoid + FB) | 3 | P1-TBDI2-27     | C1/C1S5  |
| 2x DI, 1x DO (Solenoid + FB) | 3 | P1-TBDI2-29     | C1/C1S5  |
| 2x DI, 1x DO (Solenoid + FB) | 3 | P1-TBDI2-31     | C1/C1S5  |
| 2x DI, 1x DO (Solenoid + FB) | 3 | P1-TBDI3-1      | C1/C1S6  |
| 1x AI                        | 1 | P1-TBAI1-7(A/B) | C1/C1S10 |
| 1x AI                        | 1 | P1-TBAI1-8(A/B) | C1/C1S10 |



|                   |                |                  |
|-------------------|----------------|------------------|
| TYP-DO-01 / VALVE | READY FOR TEST | Pending Sign-off |
| TYP-DO-01 / VALVE | READY FOR TEST | Pending Sign-off |
| TYP-DO-01 / VALVE | READY FOR TEST | Pending Sign-off |
| TYP-DO-01 / VALVE | READY FOR TEST | Pending Sign-off |
| TYP-DO-01 / VALVE | READY FOR TEST | Pending Sign-off |
| TYP-DO-01 / VALVE | READY FOR TEST | Pending Sign-off |
| TYP-DO-01 / VALVE | READY FOR TEST | Pending Sign-off |
| TYP-DO-01 / VALVE | READY FOR TEST | Pending Sign-off |
| TYP-DO-01 / VALVE | READY FOR TEST | Pending Sign-off |
| TYP-DO-01 / VALVE | READY FOR TEST | Pending Sign-off |
| TYP-DO-01 / VALVE | READY FOR TEST | Pending Sign-off |
| TYP-DO-01 / VALVE | READY FOR TEST | Pending Sign-off |
| TYP-DO-01 / VALVE | READY FOR TEST | Pending Sign-off |
| TYP-GEN-01        | READY FOR TEST | Pending Sign-off |
| TYP-GEN-01        | READY FOR TEST | Pending Sign-off |

|  |                                  |
|--|----------------------------------|
|  | KALASIN ENGINEERING              |
|  | Equipment Acceptance Test Report |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                 |
|----------------------|-----------------|
| Equipment Tag:       | FSL-40201       |
| Service Description: | Slurry low flow |
| Instrument Type:     | Flow Switch     |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description | Type |
|-------|--------------------------------|------|
| CH-01 | Flow Switch Alarm Status (DI)  | DI   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                  | Core-to-Core          |
|----------------------------------------|------------------------------------------------------|-----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20 \text{ M}\Omega$ @ 500VDC (1 min test)      | ____ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0 \text{ }\Omega$ end-to-end loop resistance | ____ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE            | -                     |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                 | Engraved SS           |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State        | Field Voltage / Contact   |
|-----------------------|------------------------|---------------------------|
| Discrete Field Sensor | Normal Condition       | 24.0 VDC / Contact Closed |
| Discrete Field Sensor | Trip / Alarm Condition | 0.0 VDC / Contact Open    |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                     |
|---------------------------------|--------------------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chamber fault banner / Hold permissive   |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role           | Name & Designation       | Company / Organization        |
|-------------------------|--------------------------|-------------------------------|
| Tested by (Contractor): | Lead Instrument Engineer | Kalasin Engineering Co., Ltd. |

|                                        |                                |                                |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: FSL-40201 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-01                          |
| Calibrated Range:  | 0 - 100%                                  |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| FSL-40 201_DI_1 | C1/C1S4 19           | P1-TBDI1-17    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status    | SCADA Graphic State      | Feedback Travel Angle |
|-------------------|--------------------------|-----------------------|
| Bit = 1 (Healthy) | Status Healthy (Green)   | -                     |
| Bit = 0 (Tripped) | Status Alarm (Red Blink) | -                     |

| SCADA Alarm Text                 | Priority / Class    | Sounder Horn   |
|----------------------------------|---------------------|----------------|
| FSL-40 201 SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| FSL-40 201 WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| FSL-40 201 ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict   |
|-----------|-------------|-------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED |

|       |            |                     |
|-------|------------|---------------------|
| _____ | DD-MM-2026 | ACCEPTED & APPROVED |
| _____ | DD-MM-2026 | FINAL AUTHORIZED    |

| CERTIFICATE |  |
|-------------|--|
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
|---------------------|-----------------------|---------------------------|
| C1S4-19/P1-TBDI1-17 | P1-TBDI1-17/C1S4-19   | White (Sig), Grey (24VDC) |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Response Time | Result   | Remarks                  |
|---------------|----------|--------------------------|
| < 0.1 sec     | [ ] PASS | Normal state confirmed   |
| < 0.1 sec     | [ ] PASS | Alarm latching confirmed |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed | Remarks                         |
|---------------|-------------------|---------------------------------|
| None (Clean)  | Green Loop Tag    | Ready for Process Commissioning |

|      |                           |                                          |
|------|---------------------------|------------------------------------------|
| None | <b>QC Sticker Applied</b> | <i>Satisfies Contract Scope</i>          |
| None | <b>System Cleared</b>     | <i>Authorized for Cold Water Wet Run</i> |

|  |                                  |
|--|----------------------------------|
|  | KALASIN ENGINEERING CO., LTD.    |
|  | Equipment Acceptance Test Report |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                 |
|----------------------|-----------------|
| Equipment Tag:       | FSL-40202       |
| Service Description: | Slurry low flow |
| Instrument Type:     | Flow Switch     |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description | Type |
|-------|--------------------------------|------|
| CH-01 | Flow Switch Alarm Status (DI)  | DI   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                  | Core-to-Core           |
|----------------------------------------|------------------------------------------------------|------------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20 \text{ M}\Omega$ @ 500VDC (1 min test)      | _____ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0 \text{ }\Omega$ end-to-end loop resistance | _____ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE            | -                      |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                 | Engraved SS            |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State        | Field Voltage / Contact   |
|-----------------------|------------------------|---------------------------|
| Discrete Field Sensor | Normal Condition       | 24.0 VDC / Contact Closed |
| Discrete Field Sensor | Trip / Alarm Condition | 0.0 VDC / Contact Open    |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                     |
|---------------------------------|--------------------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chamber fault banner / Hold permissive   |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role           | Name & Designation       | Company / Organization        |
|-------------------------|--------------------------|-------------------------------|
| Tested by (Contractor): | Lead Instrument Engineer | Kalasin Engineering Co., Ltd. |

|                                        |                                |                                |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: FSL-40202 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-01                          |
| Calibrated Range:  | 0 - 100%                                  |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| FSL-40 202_DI_1 | C1/C1S4 20           | P1-TBDI1-18    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status    | SCADA Graphic State      | Feedback Travel Angle |
|-------------------|--------------------------|-----------------------|
| Bit = 1 (Healthy) | Status Healthy (Green)   | -                     |
| Bit = 0 (Tripped) | Status Alarm (Red Blink) | -                     |

| SCADA Alarm Text                 | Priority / Class    | Sounder Horn   |
|----------------------------------|---------------------|----------------|
| FSL-40 202 SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| FSL-40 202 WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| FSL-40 202 ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict   |
|-----------|-------------|-------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED |

|       |            |                     |
|-------|------------|---------------------|
| _____ | DD-MM-2026 | ACCEPTED & APPROVED |
| _____ | DD-MM-2026 | FINAL AUTHORIZED    |

| CERTIFICATE |  |
|-------------|--|
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
|---------------------|-----------------------|---------------------------|
| C1S4-20/P1-TBDI1-18 | P1-TBDI1-18/C1S4-20   | White (Sig), Grey (24VDC) |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Response Time | Result   | Remarks                  |
|---------------|----------|--------------------------|
| < 0.1 sec     | [ ] PASS | Normal state confirmed   |
| < 0.1 sec     | [ ] PASS | Alarm latching confirmed |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed | Remarks                         |
|---------------|-------------------|---------------------------------|
| None (Clean)  | Green Loop Tag    | Ready for Process Commissioning |

|      |                           |                                          |
|------|---------------------------|------------------------------------------|
| None | <b>QC Sticker Applied</b> | <i>Satisfies Contract Scope</i>          |
| None | <b>System Cleared</b>     | <i>Authorized for Cold Water Wet Run</i> |

|           |  |
|-----------|--|
| KALPA     |  |
| Equipment |  |

### 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                               |
|----------------------|-----------------------------------------------|
| Equipment Tag:       | FT-40201                                      |
| Service Description: | Slurry flowrate & density check transfer line |
| Instrument Type:     | Coriolis Flow Transmitter                     |

### 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                             | Type |
|-------|------------------------------------------------------------|------|
| CH-01 | Analog Flow Rate Measurement (4-20mA AI) [Pins A+B]        | AI   |
| CH-02 | Analog Density / Temperature / Mass (4-20mA AI) [Pins A+B] | AI   |
| CH-03 | Pulse / High-Speed Batch Counter Input (DI)                | DI   |
| PWR   | Instrument Auxiliary Power Supply (24VDC / 220VAC)         | PWR  |

### 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                            | Core-to-Core           |
|----------------------------------------|------------------------------------------------|------------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test) | _____ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\ \Omega$ end-to-end loop resistance  | _____ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE      | -                      |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved           | Engraved SS            |

### 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

#### 4.1 ANALOG 5-POINT CALIBRATION TABLE (Current Loop Simulation: Pin A = IN-x, Pin B = i RTN-)

| Calibration Point | Simulated Input (mA) | Field Indicator Value |
|-------------------|----------------------|-----------------------|
| 0% (Zero)         | 4.000 mA             | 4.00 mA               |
| 25% (Span 1/4)    | 8.000 mA             | 8.01 mA               |
| 50% (Mid)         | 12.000 mA            | 12.00 mA              |
| 75% (Span 3/4)    | 16.000 mA            | 15.99 mA              |
| 100% (Full Span)  | 20.000 mA            | 20.00 mA              |

#### 4.2 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State        | Field Voltage / Contact   |
|-----------------------|------------------------|---------------------------|
| Discrete Field Sensor | Normal Condition       | 24.0 VDC / Contact Closed |
| Discrete Field Sensor | Trip / Alarm Condition | 0.0 VDC / Contact Open    |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition             | Expected Safe Action                     |
|---------------------------------|-------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal <3.6mA or >21.0mA      | Chamber fault banner / Hold non-miscible |
| SCADA Alarm Acknowledge         | Operator Acknowledge click    | Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                   | Name & Designation             | Company / Organization         |
|---------------------------------|--------------------------------|--------------------------------|
| Tested by (Contractor):         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| Witnessed by (Client QC):       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| Approved by (Commissioning PM): | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: FT-40201 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-02                          |
| Calibrated Range:  | 0-15 m³/hr                                |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name  | PLC Hardware Address      | JB Terminal ID    |
|---------------|---------------------------|-------------------|
| FT-40201_AI_1 | C1/C1S10 Pins 2(A) & 1(B) | P1-TBAI1-1(A/B)   |
| FT-40201_AI_2 | C1/C1S10 Pins 4(A) & 3(B) | P1-TBAI1-2(A/B)   |
| FT-40201_DI_1 | C1/C1S4 21                | P1-TBDI1-19       |
| FT-40201_PWR  | CA1 PSU1/2 Redundant Bus  | JB-401-TB24V / 0V |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| c)             |                    |                    |
|----------------|--------------------|--------------------|
| PLC Raw Counts | SCADA Indicated EU | Calculated Error % |
| 0 counts       | 0.0 EU             | +0.00%             |
| 8192 counts    | 25.0 EU            | +0.05%             |
| 16384 counts   | 50.0 EU            | +0.00%             |
| 24576 counts   | 74.9 EU            | -0.05%             |
| 32767 counts   | 100.0 EU           | +0.00%             |

| PLC Bit Status    | SCADA Graphic State      | Feedback Travel Angle |
|-------------------|--------------------------|-----------------------|
| Bit = 1 (Healthy) | Status Healthy (Green)   | -                     |
| Bit = 0 (Tripped) | Status Alarm (Red Blink) | -                     |

| SCADA Alarm Text                | Priority / Class    | Sounder Horn   |
|---------------------------------|---------------------|----------------|
| FT-40 201 SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| FT-40 201 WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| FT-40 201 ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| il Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side)                                   | Wire Tag (Field Side)                                 | Core Wiring Colors                    |
|-------------------------------------------------------|-------------------------------------------------------|---------------------------------------|
| C1S10-2/P1-TBAI1-1(A) (+) / C1S10-1/P1-TBAI1-1(B) (-) | P1-TBAI1-1(A)/C1S10-2 (+) / P1-TBAI1-1(B)/C1S10-1 (-) | White (+ / IN-x), Black (- / i RTN-x) |
| C1S10-4/P1-TBAI1-2(A) (+) / C1S10-3/P1-TBAI1-2(B) (-) | P1-TBAI1-2(A)/C1S10-4 (+) / P1-TBAI1-2(B)/C1S10-3 (-) | White (+ / IN-x), Black (- / i RTN-x) |
| C1S4-21/P1-TBDI1-19                                   | P1-TBDI1-19/C1S4-21                                   | White (Sig), Grey (24VDC)             |
| CA1-PSU-24V                                           | FT-40 201-V+                                          | Brown (+), Blue (-)                   |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Max Allowable Error | Result   | Remarks                         |
|---------------------|----------|---------------------------------|
| ± 0.20%             | [ ] PASS | Fluke 789 Injected Across A & B |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Span limit verified             |

| Response Time | Result   | Remarks                  |
|---------------|----------|--------------------------|
| < 0.1 sec     | [ ] PASS | Normal state confirmed   |
| < 0.1 sec     | [ ] PASS | Alarm latching confirmed |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |       |
|--|-------|
|  | KAL/  |
|  | Equip |

### 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                              |
|----------------------|----------------------------------------------|
| Equipment Tag:       | FT-40202                                     |
| Service Description: | Feed Slurry Flow Transmitter for Pipe (PIPE) |
| Instrument Type:     | Magnetic Flow Transmitter                    |

### 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                      | Type |
|-------|-----------------------------------------------------|------|
| CH-01 | Analog Flow Rate Measurement (4-20mA AI) [Pins A+B] | AI   |
| CH-02 | Pulse / High-Speed Batch Counter Input (DI)         | DI   |
| PWR   | Instrument Auxiliary Power Supply (24VDC / 220VAC)  | PWR  |

### 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

### 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

#### 4.1 ANALOG 5-POINT CALIBRATION TABLE (Current Loop Simulation: Pin A = IN-x, Pin B = i RTN->

| Calibration Point | Simulated Input (mA) | Field Indicator Value |
|-------------------|----------------------|-----------------------|
| 0% (Zero)         | 4.000 mA             | 4.00 mA               |
| 25% (Span 1/4)    | 8.000 mA             | 8.01 mA               |
| 50% (Mid)         | 12.000 mA            | 12.00 mA              |
| 75% (Span 3/4)    | 16.000 mA            | 15.99 mA              |
| 100% (Full Span)  | 20.000 mA            | 20.00 mA              |

#### 4.2 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State        | Field Voltage / Contact   |
|-----------------------|------------------------|---------------------------|
| Discrete Field Sensor | Normal Condition       | 24.0 VDC / Contact Closed |
| Discrete Field Sensor | Trip / Alarm Condition | 0.0 VDC / Contact Open    |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition             | Expected Safe Action                     |
|---------------------------------|-------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal <3.6mA or >21.0mA      | Chamber fault banner / Hold permissive   |
| SCADA Alarm Acknowledge         | Operator Acknowledge click    | Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                   | Name & Designation             | Company / Organization         |
|---------------------------------|--------------------------------|--------------------------------|
| Tested by (Contractor):         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| Witnessed by (Client QC):       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| Approved by (Commissioning PM): | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: FT-40202 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-02                          |
| Calibrated Range:  | 0 to 8,000 m³/hr                          |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name  | PLC Hardware Address      | JB Terminal ID    |
|---------------|---------------------------|-------------------|
| FT-40202_AI_1 | C1/C1S10 Pins 6(A) & 5(B) | P1-TBAI1-3(A/B)   |
| FT-40202_DI_1 | C1/C1S4 22                | P1-TBDI1-20       |
| FT-40202_PWR  | CA1 PSU1/2 Redundant Bus  | JB-401-TB24V / 0V |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| c)             |                    |                    |
|----------------|--------------------|--------------------|
| PLC Raw Counts | SCADA Indicated EU | Calculated Error % |
| 0 counts       | 0.0 EU             | +0.00%             |
| 8192 counts    | 25.0 EU            | +0.05%             |
| 16384 counts   | 50.0 EU            | +0.00%             |
| 24576 counts   | 74.9 EU            | -0.05%             |
| 32767 counts   | 100.0 EU           | +0.00%             |

| PLC Bit Status    | SCADA Graphic State      | Feedback Travel Angle |
|-------------------|--------------------------|-----------------------|
| Bit = 1 (Healthy) | Status Healthy (Green)   | -                     |
| Bit = 0 (Tripped) | Status Alarm (Red Blink) | -                     |

| SCADA Alarm Text                | Priority / Class    | Sounder Horn   |
|---------------------------------|---------------------|----------------|
| FT-40 202 SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| FT-40 202 WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| FT-40 202 ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

**CERTIFICATE**

(Official Channel)

Channel)

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 2 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side)                                   | Wire Tag (Field Side)                                 | Core Wiring Colors                    |
|-------------------------------------------------------|-------------------------------------------------------|---------------------------------------|
| C1S10-6/P1-TBAI1-3(A) (+) / C1S10-5/P1-TBAI1-3(B) (-) | P1-TBAI1-3(A)/C1S10-6 (+) / P1-TBAI1-3(B)/C1S10-5 (-) | White (+ / IN-x), Black (- / i RTN-x) |
| C1S4-22/P1-TBDI1-20                                   | P1-TBDI1-20/C1S4-22                                   | White (Sig), Grey (24VDC)             |
| CA1-PSU-24V                                           | FT-40 202-V+                                          | Brown (+), Blue (-)                   |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Max Allowable Error | Result   | Remarks                         |
|---------------------|----------|---------------------------------|
| ± 0.20%             | [ ] PASS | Fluke 789 Injected Across A & B |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Span limit verified             |

| Response Time | Result   | Remarks                  |
|---------------|----------|--------------------------|
| < 0.1 sec     | [ ] PASS | Normal state confirmed   |
| < 0.1 sec     | [ ] PASS | Alarm latching confirmed |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |       |
|--|-------|
|  | KAL   |
|  | Equip |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                            |
|----------------------|----------------------------|
| Equipment Tag:       | HV-40201                   |
| Service Description: | Slurry Drain line          |
| Instrument Type:     | Valve With Position Switch |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description               | Type |
|-------|----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI) | DI   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                  | Core-to-Core         |
|----------------------------------------|------------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20 \text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0 \text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE            | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                 | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                     |
|---------------------------------|--------------------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chamber fault banner / Hold permissive   |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role | Name & Designation | Company / Organization |
|---------------|--------------------|------------------------|
|---------------|--------------------|------------------------|

|                                        |                                |                                |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: HV-40201 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-01                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name   | PLC Hardware Address | JB Terminal ID |
|----------------|----------------------|----------------|
| HV-40 201_DI_1 | C1/C1S4 23           | P1-TBDI1-21    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text             | Priority / Class    | Sounder Horn   |
|------------------------------|---------------------|----------------|
| HV-40 201 SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| HV-40 201 WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| HV-40 201 ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict |
|-----------|-------------|-----------------|
|-----------|-------------|-----------------|

|       |            |                     |
|-------|------------|---------------------|
| _____ | DD-MM-2026 | LOOP CHECK PASSED   |
| _____ | DD-MM-2026 | ACCEPTED & APPROVED |
| _____ | DD-MM-2026 | FINAL AUTHORIZED    |

| CERTIFICATE |  |
|-------------|--|
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
|---------------------|-----------------------|---------------------------|
| C1S4-23/P1-TBDI1-21 | P1-TBDI1-21/C1S4-23   | White (Sig), Grey (24VDC) |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Response Time | Result   | Remarks                       |
|---------------|----------|-------------------------------|
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed | Remarks |
|---------------|-------------------|---------|

|              |                           |                                          |
|--------------|---------------------------|------------------------------------------|
| None (Clean) | <b>Green Loop Tag</b>     | <i>Ready for Process Commissioning</i>   |
| None         | <b>QC Sticker Applied</b> | <i>Satisfies Contract Scope</i>          |
| None         | <b>System Cleared</b>     | <i>Authorized for Cold Water Wet Run</i> |

|  |       |
|--|-------|
|  | KAL   |
|  | Equip |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                            |
|----------------------|----------------------------|
| Equipment Tag:       | HV-40202                   |
| Service Description: | Slurry Drain line          |
| Instrument Type:     | Valve With Position Switch |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description               | Type |
|-------|----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI) | DI   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                  | Core-to-Core         |
|----------------------------------------|------------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20 \text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0 \text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE            | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                 | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                     |
|---------------------------------|--------------------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chamber fault banner / Hold permissive   |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role | Name & Designation | Company / Organization |
|---------------|--------------------|------------------------|
|---------------|--------------------|------------------------|

|                                        |                                |                                |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: HV-40202 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-01                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

|                |                      |                |
|----------------|----------------------|----------------|
| PLC Tag Name   | PLC Hardware Address | JB Terminal ID |
| HV-40 202_DI_1 | C1/C1S4 24           | P1-TBDI1-22    |

|                |               |                 |
|----------------|---------------|-----------------|
| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

|                        |                       |                       |
|------------------------|-----------------------|-----------------------|
| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

|                              |                     |                |
|------------------------------|---------------------|----------------|
| SCADA Alarm Text             | Priority / Class    | Sounder Horn   |
| HV-40 202 SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| HV-40 202 WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| HV-40 202 ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

|           |             |                 |
|-----------|-------------|-----------------|
| Signature | Date Signed | Overall Verdict |
|-----------|-------------|-----------------|

|       |            |                     |
|-------|------------|---------------------|
| _____ | DD-MM-2026 | LOOP CHECK PASSED   |
| _____ | DD-MM-2026 | ACCEPTED & APPROVED |
| _____ | DD-MM-2026 | FINAL AUTHORIZED    |

| CERTIFICATE |  |
|-------------|--|
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
|---------------------|-----------------------|---------------------------|
| C1S4-24/P1-TBDI1-22 | P1-TBDI1-22/C1S4-24   | White (Sig), Grey (24VDC) |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Response Time | Result   | Remarks                       |
|---------------|----------|-------------------------------|
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed | Remarks |
|---------------|-------------------|---------|

|              |                           |                                          |
|--------------|---------------------------|------------------------------------------|
| None (Clean) | <b>Green Loop Tag</b>     | <i>Ready for Process Commissioning</i>   |
| None         | <b>QC Sticker Applied</b> | <i>Satisfies Contract Scope</i>          |
| None         | <b>System Cleared</b>     | <i>Authorized for Cold Water Wet Run</i> |

|  |       |
|--|-------|
|  | KAL/  |
|  | Equip |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                            |
|----------------------|----------------------------|
| Equipment Tag:       | HV-40203                   |
| Service Description: | Slurry Drain tank TS-40201 |
| Instrument Type:     | Valve With Position Switch |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description               | Type |
|-------|----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI) | DI   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                  | Core-to-Core         |
|----------------------------------------|------------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20 \text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0 \text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE            | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                 | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                     |
|---------------------------------|--------------------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chamber fault banner / Hold permissive   |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role | Name & Designation | Company / Organization |
|---------------|--------------------|------------------------|
|---------------|--------------------|------------------------|

|                                        |                                |                                |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: HV-40203 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-01                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

|                |                      |                |
|----------------|----------------------|----------------|
| PLC Tag Name   | PLC Hardware Address | JB Terminal ID |
| HV-40 203_DI_1 | C1/C1S4 25           | P1-TBDI1-23    |

|                |               |                 |
|----------------|---------------|-----------------|
| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

|                        |                       |                       |
|------------------------|-----------------------|-----------------------|
| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

|                              |                     |                |
|------------------------------|---------------------|----------------|
| SCADA Alarm Text             | Priority / Class    | Sounder Horn   |
| HV-40 203 SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| HV-40 203 WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| HV-40 203 ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

|           |             |                 |
|-----------|-------------|-----------------|
| Signature | Date Signed | Overall Verdict |
|-----------|-------------|-----------------|

|       |            |                     |
|-------|------------|---------------------|
| _____ | DD-MM-2026 | LOOP CHECK PASSED   |
| _____ | DD-MM-2026 | ACCEPTED & APPROVED |
| _____ | DD-MM-2026 | FINAL AUTHORIZED    |

| CERTIFICATE |  |
|-------------|--|
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
|---------------------|-----------------------|---------------------------|
| C1S4-25/P1-TBDI1-23 | P1-TBDI1-23/C1S4-25   | White (Sig), Grey (24VDC) |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Response Time | Result   | Remarks                       |
|---------------|----------|-------------------------------|
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed | Remarks |
|---------------|-------------------|---------|

|              |                           |                                          |
|--------------|---------------------------|------------------------------------------|
| None (Clean) | <b>Green Loop Tag</b>     | <i>Ready for Process Commissioning</i>   |
| None         | <b>QC Sticker Applied</b> | <i>Satisfies Contract Scope</i>          |
| None         | <b>System Cleared</b>     | <i>Authorized for Cold Water Wet Run</i> |

|  |       |
|--|-------|
|  | KAL/  |
|  | Equip |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                            |
|----------------------|----------------------------|
| Equipment Tag:       | HV-40204                   |
| Service Description: | Slurry Drain tank TS-40202 |
| Instrument Type:     | Valve With Position Switch |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description               | Type |
|-------|----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI) | DI   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                  | Core-to-Core         |
|----------------------------------------|------------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20 \text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0 \text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE            | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                 | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Alarm fault banner / Hold                           |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role | Name & Designation | Company / Organization |
|---------------|--------------------|------------------------|
|---------------|--------------------|------------------------|

|                                        |                                |                                |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: HV-40204 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-01                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

|                |                      |                |
|----------------|----------------------|----------------|
| PLC Tag Name   | PLC Hardware Address | JB Terminal ID |
| HV-40 204_DI_1 | C1/C1S4 26           | P1-TBDI1-24    |

|                |               |                 |
|----------------|---------------|-----------------|
| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

|                        |                       |                       |
|------------------------|-----------------------|-----------------------|
| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

|                                 |                     |                |
|---------------------------------|---------------------|----------------|
| SCADA Alarm Text                | Priority / Class    | Sounder Horn   |
| HV-40 204 SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| HV-40 204 WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| HV-40 204 ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

|           |             |                 |
|-----------|-------------|-----------------|
| Signature | Date Signed | Overall Verdict |
|-----------|-------------|-----------------|

|       |            |                     |
|-------|------------|---------------------|
| _____ | DD-MM-2026 | LOOP CHECK PASSED   |
| _____ | DD-MM-2026 | ACCEPTED & APPROVED |
| _____ | DD-MM-2026 | FINAL AUTHORIZED    |

| CERTIFICATE |  |
|-------------|--|
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
|---------------------|-----------------------|---------------------------|
| C1S4-26/P1-TBDI1-24 | P1-TBDI1-24/C1S4-26   | White (Sig), Grey (24VDC) |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Response Time | Result   | Remarks                       |
|---------------|----------|-------------------------------|
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed | Remarks |
|---------------|-------------------|---------|

|              |                           |                                          |
|--------------|---------------------------|------------------------------------------|
| None (Clean) | <b>Green Loop Tag</b>     | <i>Ready for Process Commissioning</i>   |
| None         | <b>QC Sticker Applied</b> | <i>Satisfies Contract Scope</i>          |
| None         | <b>System Cleared</b>     | <i>Authorized for Cold Water Wet Run</i> |

|  |       |
|--|-------|
|  | KAL/  |
|  | Equip |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                            |
|----------------------|----------------------------|
| Equipment Tag:       | HV-40205                   |
| Service Description: | Slurry Drain line          |
| Instrument Type:     | Valve With Position Switch |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description               | Type |
|-------|----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI) | DI   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                  | Core-to-Core         |
|----------------------------------------|------------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20 \text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0 \text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE            | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                 | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                     |
|---------------------------------|--------------------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chamber fault banner / Hold permissive   |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role | Name & Designation | Company / Organization |
|---------------|--------------------|------------------------|
|---------------|--------------------|------------------------|

|                                        |                                |                                |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: HV-40205 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-01                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

|                |                      |                |
|----------------|----------------------|----------------|
| PLC Tag Name   | PLC Hardware Address | JB Terminal ID |
| HV-40 205_DI_1 | C1/C1S4 27           | P1-TBDI1-25    |

|                |               |                 |
|----------------|---------------|-----------------|
| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

|                        |                       |                       |
|------------------------|-----------------------|-----------------------|
| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

|                                 |                     |                |
|---------------------------------|---------------------|----------------|
| SCADA Alarm Text                | Priority / Class    | Sounder Horn   |
| HV-40 205 SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| HV-40 205 WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| HV-40 205 ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

|           |             |                 |
|-----------|-------------|-----------------|
| Signature | Date Signed | Overall Verdict |
|-----------|-------------|-----------------|

|       |            |                     |
|-------|------------|---------------------|
| _____ | DD-MM-2026 | LOOP CHECK PASSED   |
| _____ | DD-MM-2026 | ACCEPTED & APPROVED |
| _____ | DD-MM-2026 | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S4-27/P1-TBDI1-25 | P1-TBDI1-25/C1S4-27   | White (Sig), Grey (24VDC) |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

|               |                   |         |
|---------------|-------------------|---------|
|               |                   |         |
| Punchlist Ref | Tag Color Affixed | Remarks |

|              |                           |                                          |
|--------------|---------------------------|------------------------------------------|
| None (Clean) | <b>Green Loop Tag</b>     | <i>Ready for Process Commissioning</i>   |
| None         | <b>QC Sticker Applied</b> | <i>Satisfies Contract Scope</i>          |
| None         | <b>System Cleared</b>     | <i>Authorized for Cold Water Wet Run</i> |

|  |       |
|--|-------|
|  | KAL/  |
|  | Equip |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                            |
|----------------------|----------------------------|
| Equipment Tag:       | HV-40206                   |
| Service Description: | Slurry Drain tank TS-40203 |
| Instrument Type:     | Valve With Position Switch |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description               | Type |
|-------|----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI) | DI   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                  | Core-to-Core         |
|----------------------------------------|------------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20 \text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0 \text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE            | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                 | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                     |
|---------------------------------|--------------------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Alarm fault banner / Hold permissive     |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role | Name & Designation | Company / Organization |
|---------------|--------------------|------------------------|
|---------------|--------------------|------------------------|

|                                        |                                |                                |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: HV-40206 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-01                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name   | PLC Hardware Address | JB Terminal ID |
|----------------|----------------------|----------------|
| HV-40 206_DI_1 | C1/C1S4 28           | P1-TBDI1-26    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text             | Priority / Class    | Sounder Horn   |
|------------------------------|---------------------|----------------|
| HV-40 206 SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| HV-40 206 WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| HV-40 206 ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict |
|-----------|-------------|-----------------|
|-----------|-------------|-----------------|

|       |            |                     |
|-------|------------|---------------------|
| _____ | DD-MM-2026 | LOOP CHECK PASSED   |
| _____ | DD-MM-2026 | ACCEPTED & APPROVED |
| _____ | DD-MM-2026 | FINAL AUTHORIZED    |

| CERTIFICATE |  |
|-------------|--|
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
|---------------------|-----------------------|---------------------------|
| C1S4-28/P1-TBDI1-26 | P1-TBDI1-26/C1S4-28   | White (Sig), Grey (24VDC) |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Response Time | Result   | Remarks                       |
|---------------|----------|-------------------------------|
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed | Remarks |
|---------------|-------------------|---------|

|              |                           |                                          |
|--------------|---------------------------|------------------------------------------|
| None (Clean) | <b>Green Loop Tag</b>     | <i>Ready for Process Commissioning</i>   |
| None         | <b>QC Sticker Applied</b> | <i>Satisfies Contract Scope</i>          |
| None         | <b>System Cleared</b>     | <i>Authorized for Cold Water Wet Run</i> |

|  |       |
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|  | KAL   |
|  | Equip |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                              |
|----------------------|----------------------------------------------|
| Equipment Tag:       | LT-40201                                     |
| Service Description: | Level transmitter in Spray dryer infeed tank |
| Instrument Type:     | Level transmitter                            |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                  | Type |
|-------|-------------------------------------------------|------|
| CH-01 | Analog Level Transmitter (4-20mA AI) [Pins A+B] | AI   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 ANALOG 5-POINT CALIBRATION TABLE (Current Loop Simulation: Pin A = IN-x, Pin B = i RTN-)

| Calibration Point | Simulated Input (mA) | Field Indicator Value |
|-------------------|----------------------|-----------------------|
| 0% (Zero)         | 4.000 mA             | 4.00 mA               |
| 25% (Span 1/4)    | 8.000 mA             | 8.01 mA               |
| 50% (Mid)         | 12.000 mA            | 12.00 mA              |
| 75% (Span 3/4)    | 16.000 mA            | 15.99 mA              |
| 100% (Full Span)  | 20.000 mA            | 20.00 mA              |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                            | Expected Safe Action                                |
|---------------------------------|----------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip                | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{ mA}$ or $>21.0\text{ mA}$ | Channel fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                   | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: LT-40201 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-02                          |
| Calibrated Range:  | 0 - 10 m.                                 |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name  | PLC Hardware Address        | JB Terminal ID  |
|---------------|-----------------------------|-----------------|
| LT-40201_AI_1 | C1/C1S10 Pins 12(A) & 11(B) | P1-TBAI1-5(A/B) |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| c)             |                    |                    |
|----------------|--------------------|--------------------|
| PLC Raw Counts | SCADA Indicated EU | Calculated Error % |
| 0 counts       | 0.0 EU             | +0.00%             |
| 8192 counts    | 25.0 EU            | +0.05%             |
| 16384 counts   | 50.0 EU            | +0.00%             |
| 24576 counts   | 74.9 EU            | -0.05%             |
| 32767 counts   | 100.0 EU           | +0.00%             |

| SCADA Alarm Text            | Priority / Class    | Sounder Horn   |
|-----------------------------|---------------------|----------------|
| LT-40201 SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| LT-40201 WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| LT-40201 ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

## CERTIFICATE

Channel)

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side)                                        | Wire Tag (Field Side)                                      | Core Wiring Colors                    |
|------------------------------------------------------------|------------------------------------------------------------|---------------------------------------|
| C1S10-12/P1-TBAI1-5(A) (+) /<br>C1S10-11/P1-TBAI1-5(B) (-) | P1-TBAI1-5(A)/C1S10-12 (+) /<br>P1-TBAI1-5(B)/C1S10-11 (-) | White (+ / IN-x), Black (- / i RTN-x) |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Max Allowable Error | Result   | Remarks                         |
|---------------------|----------|---------------------------------|
| ± 0.20%             | [ ] PASS | Fluke 789 Injected Across A & B |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Span limit verified             |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

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| Punchlist Ref | Tag Color Affixed         | Remarks                                  |
|---------------|---------------------------|------------------------------------------|
| None (Clean)  | <b>Green Loop Tag</b>     | <i>Ready for Process Commissioning</i>   |
| None          | <b>QC Sticker Applied</b> | <i>Satisfies Contract Scope</i>          |
| None          | <b>System Cleared</b>     | <i>Authorized for Cold Water Wet Run</i> |

|             |  |
|-------------|--|
| KALAMANDIRI |  |
| Equipment   |  |

### 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                           |
|----------------------|-------------------------------------------|
| Equipment Tag:       | LT-40202                                  |
| Service Description: | Level transmitter in pH Adjustment tank 2 |
| Instrument Type:     | Level transmitter                         |

### 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                  | Type |
|-------|-------------------------------------------------|------|
| CH-01 | Analog Level Transmitter (4-20mA AI) [Pins A+B] | AI   |

### 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core          |
|----------------------------------------|-----------------------------------------------------|-----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ____ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ____ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                     |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS           |

### 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

#### 4.1 ANALOG 5-POINT CALIBRATION TABLE (Current Loop Simulation: Pin A = IN-x, Pin B = i RTN-x)

| Calibration Point | Simulated Input (mA) | Field Indicator Value |
|-------------------|----------------------|-----------------------|
| 0% (Zero)         | 4.000 mA             | 4.00 mA               |
| 25% (Span 1/4)    | 8.000 mA             | 8.01 mA               |
| 50% (Mid)         | 12.000 mA            | 12.00 mA              |
| 75% (Span 3/4)    | 16.000 mA            | 15.99 mA              |
| 100% (Full Span)  | 20.000 mA            | 20.00 mA              |

### 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                            | Expected Safe Action                                |
|---------------------------------|----------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip                | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{ mA}$ or $>21.0\text{ mA}$ | Channel fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                   | permissive Banner silences horn, latches steady red |

### 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: LT-40202 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-01                          |
| Calibrated Range:  | 0 - 10 m.                                 |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name  | PLC Hardware Address        | JB Terminal ID  |
|---------------|-----------------------------|-----------------|
| LT-40202_AI_1 | C1/C1S10 Pins 14(A) & 13(B) | P1-TBAI1-6(A/B) |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| c)             |                    |                    |
|----------------|--------------------|--------------------|
| PLC Raw Counts | SCADA Indicated EU | Calculated Error % |
| 0 counts       | 0.0 EU             | +0.00%             |
| 8192 counts    | 25.0 EU            | +0.05%             |
| 16384 counts   | 50.0 EU            | +0.00%             |
| 24576 counts   | 74.9 EU            | -0.05%             |
| 32767 counts   | 100.0 EU           | +0.00%             |

| SCADA Alarm Text            | Priority / Class    | Sounder Horn   |
|-----------------------------|---------------------|----------------|
| LT-40202 SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| LT-40202 WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| LT-40202 ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|            |  |
|------------|--|
| TIFICATE   |  |
| l Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                                                         |                                                         |                                       |
|---------------------------------------------------------|---------------------------------------------------------|---------------------------------------|
|                                                         |                                                         |                                       |
| Wire Tag (PLC Side)                                     | Wire Tag (Field Side)                                   | Core Wiring Colors                    |
| C1S10-14/P1-TBAI1-6(A) (+) / C1S10-13/P1-TBAI1-6(B) (-) | P1-TBAI1-6(A)/C1S10-14 (+) / P1-TBAI1-6(B)/C1S10-13 (-) | White (+ / IN-x), Black (- / i RTN-x) |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|                     |          |                                 |
|---------------------|----------|---------------------------------|
|                     |          |                                 |
|                     |          |                                 |
| Max Allowable Error | Result   | Remarks                         |
| ± 0.20%             | [ ] PASS | Fluke 789 Injected Across A & B |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Span limit verified             |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |       |
|--|-------|
|  | KAL   |
|  | Equip |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                              |
|----------------------|----------------------------------------------|
| Equipment Tag:       | LT-40203                                     |
| Service Description: | Level transmitter in Spray dryer infeed tank |
| Instrument Type:     | Level transmitter                            |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                  | Type |
|-------|-------------------------------------------------|------|
| CH-01 | Analog Level Transmitter (4-20mA AI) [Pins A+B] | AI   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 ANALOG 5-POINT CALIBRATION TABLE (Current Loop Simulation: Pin A = IN-x, Pin B = i RTN-)

| Calibration Point | Simulated Input (mA) | Field Indicator Value |
|-------------------|----------------------|-----------------------|
| 0% (Zero)         | 4.000 mA             | 4.00 mA               |
| 25% (Span 1/4)    | 8.000 mA             | 8.01 mA               |
| 50% (Mid)         | 12.000 mA            | 12.00 mA              |
| 75% (Span 3/4)    | 16.000 mA            | 15.99 mA              |
| 100% (Full Span)  | 20.000 mA            | 20.00 mA              |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                            | Expected Safe Action                     |
|---------------------------------|----------------------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip                | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal $<3.6\text{ mA}$ or $>21.0\text{ mA}$ | Channel fault banner / Hold permissive   |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                   | Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: LT-40203 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-02                          |
| Calibrated Range:  | 0 - 10 m.                                 |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name  | PLC Hardware Address      | JB Terminal ID  |
|---------------|---------------------------|-----------------|
| LT-40203_AI_1 | C1/C1S10 Pins 8(A) & 7(B) | P1-TBAI1-4(A/B) |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| c)             |                    |                    |
|----------------|--------------------|--------------------|
| PLC Raw Counts | SCADA Indicated EU | Calculated Error % |
| 0 counts       | 0.0 EU             | +0.00%             |
| 8192 counts    | 25.0 EU            | +0.05%             |
| 16384 counts   | 50.0 EU            | +0.00%             |
| 24576 counts   | 74.9 EU            | -0.05%             |
| 32767 counts   | 100.0 EU           | +0.00%             |

| SCADA Alarm Text            | Priority / Class    | Sounder Horn   |
|-----------------------------|---------------------|----------------|
| LT-40203 SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| LT-40203 WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| LT-40203 ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

|  |  |  |
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|  |  |  |
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| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

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| l Channel) |  |

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|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                                                       |                                                       |                                       |
|-------------------------------------------------------|-------------------------------------------------------|---------------------------------------|
|                                                       |                                                       |                                       |
| Wire Tag (PLC Side)                                   | Wire Tag (Field Side)                                 | Core Wiring Colors                    |
| C1S10-8/P1-TBAI1-4(A) (+) / C1S10-7/P1-TBAI1-4(B) (-) | P1-TBAI1-4(A)/C1S10-8 (+) / P1-TBAI1-4(B)/C1S10-7 (-) | White (+ / IN-x), Black (- / i RTN-x) |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|                     |          |                                 |
|---------------------|----------|---------------------------------|
|                     |          |                                 |
|                     |          |                                 |
| Max Allowable Error | Result   | Remarks                         |
| ± 0.20%             | [ ] PASS | Fluke 789 Injected Across A & B |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Span limit verified             |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

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| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |       |
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|  | KAL   |
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## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                                          |
|----------------------|----------------------------------------------------------|
| Equipment Tag:       | PT-40201                                                 |
| Service Description: | Pressure Transmitter for Inlet Filter Strainer (FSTR-11) |
| Instrument Type:     | Pressure Transmitter                                     |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                                      | Type |
|-------|---------------------------------------------------------------------|------|
| CH-01 | Analog Pressure / Diff. Pressure Transmitter (4-20mA AI) [Pins A+B] | AI   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                  | Core-to-Core         |
|----------------------------------------|------------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20 \text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0 \text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE            | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                 | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 ANALOG 5-POINT CALIBRATION TABLE (Current Loop Simulation: Pin A = IN-x, Pin B = i RTN-)

| Calibration Point | Simulated Input (mA) | Field Indicator Value |
|-------------------|----------------------|-----------------------|
| 0% (Zero)         | 4.000 mA             | 4.00 mA               |
| 25% (Span 1/4)    | 8.000 mA             | 8.01 mA               |
| 50% (Mid)         | 12.000 mA            | 12.00 mA              |
| 75% (Span 3/4)    | 16.000 mA            | 15.99 mA              |
| 100% (Full Span)  | 20.000 mA            | 20.00 mA              |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                     |
|---------------------------------|--------------------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Channel fault banner / Hold permissive   |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: PT-40201 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-02                          |
| Calibrated Range:  | −100 to 500 mBar                          |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name   | PLC Hardware Address        | JB Terminal ID  |
|----------------|-----------------------------|-----------------|
| PT-40 201_AI_1 | C1/C1S10 Pins 20(A) & 19(B) | P1-TBAI1-9(A/B) |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| c)             |                    |                    |
|----------------|--------------------|--------------------|
| PLC Raw Counts | SCADA Indicated EU | Calculated Error % |
| 0 counts       | 0.0 EU             | +0.00%             |
| 8192 counts    | 25.0 EU            | +0.05%             |
| 16 384 counts  | 50.0 EU            | +0.00%             |
| 24 576 counts  | 74.9 EU            | −0.05%             |
| 32 767 counts  | 100.0 EU           | +0.00%             |

| SCADA Alarm Text                | Priority / Class    | Sounder Horn   |
|---------------------------------|---------------------|----------------|
| PT-40 201 SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| PT-40 201 WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| PT-40 201 ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

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| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

## CERTIFICATE

al Channel)

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wiring Color Legend                                        |                                                            |                                       |
|------------------------------------------------------------|------------------------------------------------------------|---------------------------------------|
| Wire Tag (PLC Side)                                        | Wire Tag (Field Side)                                      | Core Wiring Colors                    |
| C1S10-20/P1-TBAI1-9(A) (+) /<br>C1S10-19/P1-TBAI1-9(B) (-) | P1-TBAI1-9(A)/C1S10-20 (+) /<br>P1-TBAI1-9(B)/C1S10-19 (-) | White (+ / IN-x), Black (- / i RTN-x) |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Max Allowable Error | Result   | Remarks                         |
|---------------------|----------|---------------------------------|
| ± 0.20%             | [ ] PASS | Fluke 789 Injected Across A & B |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Span limit verified             |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

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| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |       |
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## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                                  |
|----------------------|--------------------------------------------------|
| Equipment Tag:       | PT-40202                                         |
| Service Description: | Feed Slurry Pressure Transmitter for Pipe (PIPE) |
| Instrument Type:     | Pressure Transmitter                             |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                                      | Type |
|-------|---------------------------------------------------------------------|------|
| CH-01 | Analog Pressure / Diff. Pressure Transmitter (4-20mA AI) [Pins A+B] | AI   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                  | Core-to-Core         |
|----------------------------------------|------------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20 \text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0 \text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE            | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                 | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 ANALOG 5-POINT CALIBRATION TABLE (Current Loop Simulation: Pin A = IN-x, Pin B = i RTN-)

| Calibration Point | Simulated Input (mA) | Field Indicator Value |
|-------------------|----------------------|-----------------------|
| 0% (Zero)         | 4.000 mA             | 4.00 mA               |
| 25% (Span 1/4)    | 8.000 mA             | 8.01 mA               |
| 50% (Mid)         | 12.000 mA            | 12.00 mA              |
| 75% (Span 3/4)    | 16.000 mA            | 15.99 mA              |
| 100% (Full Span)  | 20.000 mA            | 20.00 mA              |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Channel fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: PT-40202 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-02                          |
| Calibrated Range:  | 0 - 10'barg                               |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name   | PLC Hardware Address        | JB Terminal ID   |
|----------------|-----------------------------|------------------|
| PT-40 202_AI_1 | C1/C1S10 Pins 22(A) & 21(B) | P1-TBAI1-10(A/B) |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| c)             |                    |                    |
|----------------|--------------------|--------------------|
| PLC Raw Counts | SCADA Indicated EU | Calculated Error % |
| 0 counts       | 0.0 EU             | +0.00%             |
| 8192 counts    | 25.0 EU            | +0.05%             |
| 16 384 counts  | 50.0 EU            | +0.00%             |
| 24 576 counts  | 74.9 EU            | -0.05%             |
| 32 767 counts  | 100.0 EU           | +0.00%             |

| SCADA Alarm Text                | Priority / Class    | Sounder Horn   |
|---------------------------------|---------------------|----------------|
| PT-40 202 SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| PT-40 202 WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| PT-40 202 ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

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| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

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| l Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                                                           |                                                           |                                       |
|-----------------------------------------------------------|-----------------------------------------------------------|---------------------------------------|
|                                                           |                                                           |                                       |
| Wire Tag (PLC Side)                                       | Wire Tag (Field Side)                                     | Core Wiring Colors                    |
| C1S10-22/P1-TBAI1-10(A) (+) / C1S10-21/P1-TBAI1-10(B) (-) | P1-TBAI1-10(A)/C1S10-22 (+) / P1-TBAI1-10(B)/C1S10-21 (-) | White (+ / IN-x), Black (- / i RTN-x) |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|                     |          |                                 |
|---------------------|----------|---------------------------------|
|                     |          |                                 |
|                     |          |                                 |
| Max Allowable Error | Result   | Remarks                         |
| ± 0.20%             | [ ] PASS | Fluke 789 Injected Across A & B |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Span limit verified             |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

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| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

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|  | Equip |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                                                          |
|----------------------|--------------------------------------------------------------------------|
| Equipment Tag:       | PT-60202                                                                 |
| Service Description: | Feed Slurry Pressure Transmitter for High Pressure Pump Outlet (PUPD-11) |
| Instrument Type:     | Pressure Transmitter                                                     |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                                      | Type |
|-------|---------------------------------------------------------------------|------|
| CH-01 | Analog Pressure / Diff. Pressure Transmitter (4-20mA AI) [Pins A+B] | AI   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 ANALOG 5-POINT CALIBRATION TABLE (Current Loop Simulation: Pin A = IN-x, Pin B = i RTN-x)

| Calibration Point | Simulated Input (mA) | Field Indicator Value |
|-------------------|----------------------|-----------------------|
| 0% (Zero)         | 4.000 mA             | 4.00 mA               |
| 25% (Span 1/4)    | 8.000 mA             | 8.01 mA               |
| 50% (Mid)         | 12.000 mA            | 12.00 mA              |
| 75% (Span 3/4)    | 16.000 mA            | 15.99 mA              |
| 100% (Full Span)  | 20.000 mA            | 20.00 mA              |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                            | Expected Safe Action                                |
|---------------------------------|----------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip                | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{ mA}$ or $>21.0\text{ mA}$ | Channel fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                   | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: PT-60202 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-602-PD-01                          |
| Calibrated Range:  |  | 0 - 100%                                  |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

|                |                             |                  |
|----------------|-----------------------------|------------------|
| PLC Tag Name   | PLC Hardware Address        | JB Terminal ID   |
| PT-60 202_AI_1 | C1/C1S10 Pins 24(A) & 23(B) | P1-TBAI1-11(A/B) |

|                |               |                 |
|----------------|---------------|-----------------|
| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

|                |                    |                    |
|----------------|--------------------|--------------------|
| d)             |                    |                    |
| PLC Raw Counts | SCADA Indicated EU | Calculated Error % |
| 0 counts       | 0.0 EU             | +0.00%             |
| 8192 counts    | 25.0 EU            | +0.05%             |
| 16 384 counts  | 50.0 EU            | +0.00%             |
| 24 576 counts  | 74.9 EU            | -0.05%             |
| 32 767 counts  | 100.0 EU           | +0.00%             |

|                                 |                     |                |
|---------------------------------|---------------------|----------------|
| SCADA Alarm Text                | Priority / Class    | Sounder Horn   |
| PT-60 202 SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| PT-60 202 WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| PT-60 202 ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

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| l Channel) |  |

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|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                                                           |                                                           |                                       |
|-----------------------------------------------------------|-----------------------------------------------------------|---------------------------------------|
|                                                           |                                                           |                                       |
| Wire Tag (PLC Side)                                       | Wire Tag (Field Side)                                     | Core Wiring Colors                    |
| C1S10-24/P1-TBAI1-11(A) (+) / C1S10-23/P1-TBAI1-11(B) (-) | P1-TBAI1-11(A)/C1S10-24 (+) / P1-TBAI1-11(B)/C1S10-23 (-) | White (+ / IN-x), Black (- / i RTN-x) |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|                     |          |                                 |
|---------------------|----------|---------------------------------|
|                     |          |                                 |
|                     |          |                                 |
| Max Allowable Error | Result   | Remarks                         |
| ± 0.20%             | [ ] PASS | Fluke 789 Injected Across A & B |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Span limit verified             |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|           |  |
|-----------|--|
| KALAHARI  |  |
| Equipment |  |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                                   |
|----------------------|---------------------------------------------------|
| Equipment Tag:       | SV-40201                                          |
| Service Description: | Water service pH analyzer in pH Adjustment tank 1 |
| Instrument Type:     | Solenoid valve                                    |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description              | Type |
|-------|---------------------------------------------|------|
| CH-01 | Solenoid Valve Actuation Command (24VDC DO) | DO   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                            | Core-to-Core          |
|----------------------------------------|------------------------------------------------|-----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test) | ____ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\ \Omega$ end-to-end loop resistance  | ____ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE      | -                     |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved           | Engraved SS           |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                     |
|---------------------------------|--------------------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chamber fault banner / Hold permissive   |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role | Name & Designation | Company / Organization |
|---------------|--------------------|------------------------|
|---------------|--------------------|------------------------|

|                                        |                                |                                |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: SV-40201 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-01                          |
| Calibrated Range:  | On-Off                                    |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name  | PLC Hardware Address | JB Terminal ID |
|---------------|----------------------|----------------|
| SV-40201_DO_1 | C1/C1S8 27           | P1-TBRL1-25    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text               | Priority / Class    | Sounder Horn   |
|--------------------------------|---------------------|----------------|
| SV-40201 SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| SV-40201 WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| SV-40201 ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict |
|-----------|-------------|-----------------|

|       |            |                     |
|-------|------------|---------------------|
| _____ | DD-MM-2026 | LOOP CHECK PASSED   |
| _____ | DD-MM-2026 | ACCEPTED & APPROVED |
| _____ | DD-MM-2026 | FINAL AUTHORIZED    |

## CERTIFICATE

al Channel)

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors      |
|---------------------|-----------------------|-------------------------|
| C1S8-27/P1-TBRL1-25 | P1-TBRL1-25/C1S8-27   | 24VDC Red (+), Blue (-) |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Response Time | Result   | Remarks                       |
|---------------|----------|-------------------------------|
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist     |                   |         |
|---------------|-------------------|---------|
| Punchlist Ref | Tag Color Affixed | Remarks |
|               |                   |         |

|              |                           |                                          |
|--------------|---------------------------|------------------------------------------|
| None (Clean) | <b>Green Loop Tag</b>     | <i>Ready for Process Commissioning</i>   |
| None         | <b>QC Sticker Applied</b> | <i>Satisfies Contract Scope</i>          |
| None         | <b>System Cleared</b>     | <i>Authorized for Cold Water Wet Run</i> |

|           |  |
|-----------|--|
| KALAHARI  |  |
| Equipment |  |

### 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                                   |
|----------------------|---------------------------------------------------|
| Equipment Tag:       | SV-40202                                          |
| Service Description: | Water service pH analyzer in pH Adjustment tank 2 |
| Instrument Type:     | Solenoid valve                                    |

### 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description              | Type |
|-------|---------------------------------------------|------|
| CH-01 | Solenoid Valve Actuation Command (24VDC DO) | DO   |

### 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core          |
|----------------------------------------|-----------------------------------------------------|-----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ____ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ____ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                     |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS           |

### 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

#### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

### 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                     |
|---------------------------------|--------------------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chamber fault banner / Hold permissive   |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | Banner silences horn, latches steady red |

### 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role | Name & Designation | Company / Organization |
|---------------|--------------------|------------------------|
|---------------|--------------------|------------------------|

|                                        |                                |                                |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: SV-40202 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-01                          |
| Calibrated Range:  | On-Off                                    |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name  | PLC Hardware Address | JB Terminal ID |
|---------------|----------------------|----------------|
| SV-40202_DO_1 | C1/C1S8 28           | P1-TBRL1-26    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text               | Priority / Class    | Sounder Horn   |
|--------------------------------|---------------------|----------------|
| SV-40202 SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| SV-40202 WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| SV-40202 ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict |
|-----------|-------------|-----------------|

|       |            |                     |
|-------|------------|---------------------|
| _____ | DD-MM-2026 | LOOP CHECK PASSED   |
| _____ | DD-MM-2026 | ACCEPTED & APPROVED |
| _____ | DD-MM-2026 | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                         |
|---------------------|-----------------------|-------------------------|
|                     |                       |                         |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors      |
| C1S8-28/P1-TBRL1-26 | P1-TBRL1-26/C1S8-28   | 24VDC Red (+), Blue (-) |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

|               |                   |         |
|---------------|-------------------|---------|
|               |                   |         |
| Punchlist Ref | Tag Color Affixed | Remarks |

|              |                           |                                          |
|--------------|---------------------------|------------------------------------------|
| None (Clean) | <b>Green Loop Tag</b>     | <i>Ready for Process Commissioning</i>   |
| None         | <b>QC Sticker Applied</b> | <i>Satisfies Contract Scope</i>          |
| None         | <b>System Cleared</b>     | <i>Authorized for Cold Water Wet Run</i> |

|  |       |
|--|-------|
|  | KAL/  |
|  | Equip |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                                 |
|----------------------|-------------------------------------------------|
| Equipment Tag:       | SV-40203                                        |
| Service Description: | Air service pH analyzer in pH Adjustment tank 1 |
| Instrument Type:     | Solenoid valve                                  |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description              | Type |
|-------|---------------------------------------------|------|
| CH-01 | Solenoid Valve Actuation Command (24VDC DO) | DO   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                  | Core-to-Core         |
|----------------------------------------|------------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20 \text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0 \text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE            | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                 | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                     |
|---------------------------------|--------------------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chamber fault banner / Hold permissive   |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role | Name & Designation | Company / Organization |
|---------------|--------------------|------------------------|
|---------------|--------------------|------------------------|

|                                        |                                |                                |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: SV-40203 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-01                          |
| Calibrated Range:  | On-Off                                    |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name  | PLC Hardware Address | JB Terminal ID |
|---------------|----------------------|----------------|
| SV-40203_DO_1 | C1/C1S8 29           | P1-TBRL1-27    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text            | Priority / Class    | Sounder Horn   |
|-----------------------------|---------------------|----------------|
| SV-40203 SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| SV-40203 WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| SV-40203 ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict |
|-----------|-------------|-----------------|

|       |            |                     |
|-------|------------|---------------------|
| _____ | DD-MM-2026 | LOOP CHECK PASSED   |
| _____ | DD-MM-2026 | ACCEPTED & APPROVED |
| _____ | DD-MM-2026 | FINAL AUTHORIZED    |

## CERTIFICATE

al Channel)

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors      |
|---------------------|-----------------------|-------------------------|
| C1S8-29/P1-TBRL1-27 | P1-TBRL1-27/C1S8-29   | 24VDC Red (+), Blue (-) |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Response Time | Result   | Remarks                       |
|---------------|----------|-------------------------------|
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist     |                   |         |
|---------------|-------------------|---------|
| Punchlist Ref | Tag Color Affixed | Remarks |
|               |                   |         |

|              |                           |                                          |
|--------------|---------------------------|------------------------------------------|
| None (Clean) | <b>Green Loop Tag</b>     | <i>Ready for Process Commissioning</i>   |
| None         | <b>QC Sticker Applied</b> | <i>Satisfies Contract Scope</i>          |
| None         | <b>System Cleared</b>     | <i>Authorized for Cold Water Wet Run</i> |

|  |       |
|--|-------|
|  | KAL/  |
|  | Equip |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                                 |
|----------------------|-------------------------------------------------|
| Equipment Tag:       | SV-40204                                        |
| Service Description: | Air service pH analyzer in pH Adjustment tank 2 |
| Instrument Type:     | Solenoid valve                                  |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description              | Type |
|-------|---------------------------------------------|------|
| CH-01 | Solenoid Valve Actuation Command (24VDC DO) | DO   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                  | Core-to-Core         |
|----------------------------------------|------------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20 \text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0 \text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE            | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                 | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                     |
|---------------------------------|--------------------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chamber fault banner / Hold permissive   |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role | Name & Designation | Company / Organization |
|---------------|--------------------|------------------------|
|---------------|--------------------|------------------------|

|                                        |                                |                                |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Document: SV-40204 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-01                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name   | PLC Hardware Address | JB Terminal ID |
|----------------|----------------------|----------------|
| SV-40 204_DO_1 | C1/C1S8 30           | P1-TBRL1-28    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text                | Priority / Class    | Sounder Horn   |
|---------------------------------|---------------------|----------------|
| SV-40 204 SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| SV-40 204 WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| SV-40 204 ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict |
|-----------|-------------|-----------------|
|-----------|-------------|-----------------|

|       |            |                     |
|-------|------------|---------------------|
| _____ | DD-MM-2026 | LOOP CHECK PASSED   |
| _____ | DD-MM-2026 | ACCEPTED & APPROVED |
| _____ | DD-MM-2026 | FINAL AUTHORIZED    |

**CERTIFICATE**

(Official Channel)

al Channel)

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors      |
|---------------------|-----------------------|-------------------------|
| C1S8-30/P1-TBRL1-28 | P1-TBRL1-28/C1S8-30   | 24VDC Red (+), Blue (-) |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Response Time | Result   | Remarks                       |
|---------------|----------|-------------------------------|
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed | Remarks |
|---------------|-------------------|---------|

|              |                           |                                          |
|--------------|---------------------------|------------------------------------------|
| None (Clean) | <b>Green Loop Tag</b>     | <i>Ready for Process Commissioning</i>   |
| None         | <b>QC Sticker Applied</b> | <i>Satisfies Contract Scope</i>          |
| None         | <b>System Cleared</b>     | <i>Authorized for Cold Water Wet Run</i> |

|  |        |
|--|--------|
|  | KAL/   |
|  | Equip. |

### 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                     |
|----------------------|-------------------------------------|
| Equipment Tag:       | XV-40201A                           |
| Service Description: | Slurry feed-in pH Adjustment tank 1 |
| Instrument Type:     | Automatic on/off valve              |

### 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

### 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                            | Core-to-Core         |
|----------------------------------------|------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test) | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\ \Omega$ end-to-end loop resistance  | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE      | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved           | Engraved SS          |

### 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

#### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

### 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40201A | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-01                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 201A_DI_1 | C1/C1S4 29           | P1-TBDI1-27    |
| XV-40 201A_DI_2 | C1/C1S4 30           | P1-TBDI1-28    |
| XV-40 201A_DO_1 | C1/C1S8 31           | P1-TBRL1-29    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text              | Priority / Class    | Sounder Horn   |
|-------------------------------|---------------------|----------------|
| XV-40 201A SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| XV-40 201A WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| XV-40 201A ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S4-29/P1-TBDI1-27 | P1-TBDI1-27/C1S4-29   | White (Sig), Grey (24VDC) |
| C1S4-30/P1-TBDI1-28 | P1-TBDI1-28/C1S4-30   | White (Sig), Grey (24VDC) |
| C1S8-31/P1-TBRL1-29 | P1-TBRL1-29/C1S8-31   | 24VDC Red (+), Blue (-)   |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |       |
|--|-------|
|  | KAL   |
|  | Equip |

### 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                         |
|----------------------|-----------------------------------------|
| Equipment Tag:       | XV-40201B                               |
| Service Description: | Slurry feed-out Spray dryer infeed tank |
| Instrument Type:     | Automatic on/off valve                  |

### 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

### 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

### 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

#### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

### 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40201B | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-02                          |
| Calibrated Range:  | On-Off                                    |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 201B_DI_1 | C1/C1S4 31           | P1-TBDI1-29    |
| XV-40 201B_DI_2 | C1/C1S4 32           | P1-TBDI1-30    |
| XV-40 201B_DO_1 | C1/C1S8 32           | P1-TBRL1-30    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text                 | Priority / Class    | Sounder Horn   |
|----------------------------------|---------------------|----------------|
| XV-40 201B SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| XV-40 201B WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| XV-40 201B ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S4-31/P1-TBDI1-29 | P1-TBDI1-29/C1S4-31   | White (Sig), Grey (24VDC) |
| C1S4-32/P1-TBDI1-30 | P1-TBDI1-30/C1S4-32   | White (Sig), Grey (24VDC) |
| C1S8-32/P1-TBRL1-30 | P1-TBRL1-30/C1S8-32   | 24VDC Red (+), Blue (-)   |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |       |
|--|-------|
|  | KAL   |
|  | Equip |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                                   |
|----------------------|---------------------------------------------------|
| Equipment Tag:       | XV-40201C                                         |
| Service Description: | Water washing line outlet Spray dryer infeed tank |
| Instrument Type:     | Automatic on/off valve                            |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                            | Core-to-Core         |
|----------------------------------------|------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test) | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\ \Omega$ end-to-end loop resistance  | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE      | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved           | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40201C | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-02                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 201C_DI_1 | C1/C1S4 33           | P1-TBDI1-31    |
| XV-40 201C_DI_2 | C1/C1S4 34           | P1-TBDI1-32    |
| XV-40 201C_DO_1 | C1/C1S8 33           | P1-TBRL1-31    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text              | Priority / Class    | Sounder Horn   |
|-------------------------------|---------------------|----------------|
| XV-40 201C SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| XV-40 201C WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| XV-40 201C ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S4-33/P1-TBDI1-31 | P1-TBDI1-31/C1S4-33   | White (Sig), Grey (24VDC) |
| C1S4-34/P1-TBDI1-32 | P1-TBDI1-32/C1S4-34   | White (Sig), Grey (24VDC) |
| C1S8-33/P1-TBRL1-31 | P1-TBRL1-31/C1S8-33   | 24VDC Red (+), Blue (-)   |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |        |
|--|--------|
|  | KAL    |
|  | Equip. |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                           |
|----------------------|-------------------------------------------|
| Equipment Tag:       | XV-40201D                                 |
| Service Description: | CIP water washing Spray dryer infeed tank |
| Instrument Type:     | Automatic on/off valve                    |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40201D | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-02                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 201D_DI_1 | C1/C1S5 1            | P1-TBDI2-1     |
| XV-40 201D_DI_2 | C1/C1S5 2            | P1-TBDI2-2     |
| XV-40 201D_DO_1 | C1/C1S8 34           | P1-TBRL1-32    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text                 | Priority / Class    | Sounder Horn   |
|----------------------------------|---------------------|----------------|
| XV-40 201D SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| XV-40 201D WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| XV-40 201D ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

| CERTIFICATE |  |
|-------------|--|
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
|---------------------|-----------------------|---------------------------|
| C1S5-1/P1-TBDI2-1   | P1-TBDI2-1/C1S5-1     | White (Sig), Grey (24VDC) |
| C1S5-2/P1-TBDI2-2   | P1-TBDI2-2/C1S5-2     | White (Sig), Grey (24VDC) |
| C1S8-34/P1-TBRL1-32 | P1-TBRL1-32/C1S8-34   | 24VDC Red (+), Blue (-)   |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Response Time | Result   | Remarks                       |
|---------------|----------|-------------------------------|
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

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|  | Equip |

### 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                                          |
|----------------------|----------------------------------------------------------|
| Equipment Tag:       | XV-40201E                                                |
| Service Description: | CIP hight pressure water washing Spray dryer infeed tank |
| Instrument Type:     | Automatic on/off valve                                   |

### 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

### 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

### 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

#### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

### 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40201E | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-02                          |
| Calibrated Range:  | On-Off                                    |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 201E_DI_1 | C1/C1S5 3            | P1-TBDI2-3     |
| XV-40 201E_DI_2 | C1/C1S5 4            | P1-TBDI2-4     |
| XV-40 201E_DO_1 | C1/C1S9 1            | P1-TBRL2-1     |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| ____ MΩ        | ____ MΩ       | ____ MΩ         |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text                 | Priority / Class    | Sounder Horn   |
|----------------------------------|---------------------|----------------|
| XV-40 201E SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| XV-40 201E WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| XV-40 201E ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S5-3/P1-TBDI2-3   | P1-TBDI2-3/C1S5-3     | White (Sig), Grey (24VDC) |
| C1S5-4/P1-TBDI2-4   | P1-TBDI2-4/C1S5-4     | White (Sig), Grey (24VDC) |
| C1S9-1/P1-TBRL2-1   | P1-TBRL2-1/C1S9-1     | 24VDC Red (+), Blue (-)   |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |       |
|--|-------|
|  | KAL/  |
|  | Equip |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                               |
|----------------------|-------------------------------|
| Equipment Tag:       | XV-40201F                     |
| Service Description: | Slurry feed-to spray dry tank |
| Instrument Type:     | Automatic on/off valve        |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                     |
|---------------------------------|--------------------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chamber fault banner / Hold permissive   |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40201F | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-01                          |
| Calibrated Range:  | On-Off                                    |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 201F_DI_1 | C1/C1S5 5            | P1-TBDI2-5     |
| XV-40 201F_DI_2 | C1/C1S5 6            | P1-TBDI2-6     |
| XV-40 201F_DO_1 | C1/C1S9 2            | P1-TBRL2-2     |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text                 | Priority / Class    | Sounder Horn   |
|----------------------------------|---------------------|----------------|
| XV-40 201F SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| XV-40 201F WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| XV-40 201F ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S5-5/P1-TBDI2-5   | P1-TBDI2-5/C1S5-5     | White (Sig), Grey (24VDC) |
| C1S5-6/P1-TBDI2-6   | P1-TBDI2-6/C1S5-6     | White (Sig), Grey (24VDC) |
| C1S9-2/P1-TBRL2-2   | P1-TBRL2-2/C1S9-2     | 24VDC Red (+), Blue (-)   |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |        |
|--|--------|
|  | KAL/   |
|  | Equip. |

### 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                        |
|----------------------|----------------------------------------|
| Equipment Tag:       | XV-40201G                              |
| Service Description: | CIP water washing pH Adjustment tank 1 |
| Instrument Type:     | Automatic on/off valve                 |

### 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

### 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

### 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

#### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

### 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40201G | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-01                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 201G_DI_1 | C1/C1S5 7            | P1-TBDI2-7     |
| XV-40 201G_DI_2 | C1/C1S5 8            | P1-TBDI2-8     |
| XV-40 201G_DO_1 | C1/C1S9 3            | P1-TBRL2-3     |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text              | Priority / Class    | Sounder Horn   |
|-------------------------------|---------------------|----------------|
| XV-40 201G SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| XV-40 201G WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| XV-40 201G ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
|---------------------|-----------------------|---------------------------|
| C1S5-7/P1-TBDI2-7   | P1-TBDI2-7/C1S5-7     | White (Sig), Grey (24VDC) |
| C1S5-8/P1-TBDI2-8   | P1-TBDI2-8/C1S5-8     | White (Sig), Grey (24VDC) |
| C1S9-3/P1-TBRL2-3   | P1-TBRL2-3/C1S9-3     | 24VDC Red (+), Blue (-)   |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Response Time | Result   | Remarks                       |
|---------------|----------|-------------------------------|
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |       |
|--|-------|
|  | KAL   |
|  | Equip |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                                       |
|----------------------|-------------------------------------------------------|
| Equipment Tag:       | XV-40201H                                             |
| Service Description: | CIP Hight pressure water washing pH Adjustment tank 1 |
| Instrument Type:     | Automatic on/off valve                                |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40201H | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Instrument Identification |                                           |
| P&ID Number:              | 325-01-402-PD-01                          |
| Calibrated Range:         | On-Off                                    |
| Branch Cable Spec:        | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Addressing & Terminal Mapping |                      |                |
|-----------------------------------|----------------------|----------------|
| PLC Tag Name                      | PLC Hardware Address | JB Terminal ID |
| XV-40 201H_DI_1                   | C1/C1S5 9            | P1-TBDI2-9     |
| XV-40 201H_DI_2                   | C1/C1S5 10           | P1-TBDI2-10    |
| XV-40 201H_DO_1                   | C1/C1S9 4            | P1-TBRL2-4     |

| Isolation & Safety Checks |               |                 |
|---------------------------|---------------|-----------------|
| Core-to-Shield            | Core-to-Earth | Shield-to-Earth |
| _____ MΩ                  | _____ MΩ      | _____ MΩ        |
| -                         | -             | -               |
| -                         | -             | Isolated        |
| Gland Tight               | Drain Plug OK | O-ring Intact   |

| Operational Feedback & Status |                       |                       |
|-------------------------------|-----------------------|-----------------------|
| PLC Bit Status                | SCADA Graphic State   | Feedback Travel Angle |
| Bit = 1 (Active)              | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)              | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized)        | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)              | Limit Switch Icon Lit | Close Target Hit      |

| Alarm & Safety System Configuration |                     |                |
|-------------------------------------|---------------------|----------------|
| SCADA Alarm Text                    | Priority / Class    | Sounder Horn   |
| XV-40 201H SAFETY SHUTDOWN          | High (Priority 1)   | Audible Active |
| XV-40 201H WIRE BREAK / FAULT       | Medium (Priority 2) | Warning Tone   |
| XV-40 201H ALARM ACKNOWLEDGED       | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S5-9/P1-TBDI2-9   | P1-TBDI2-9/C1S5-9     | White (Sig), Grey (24VDC) |
| C1S5-10/P1-TBDI2-10 | P1-TBDI2-10/C1S5-10   | White (Sig), Grey (24VDC) |
| C1S9-4/P1-TBRL2-4   | P1-TBRL2-4/C1S9-4     | 24VDC Red (+), Blue (-)   |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

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|  | Equip. |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                     |
|----------------------|-------------------------------------|
| Equipment Tag:       | XV-40202A                           |
| Service Description: | Slurry feed-in pH Adjustment tank 2 |
| Instrument Type:     | Automatic on/off valve              |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40202A | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-01                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 202A_DI_1 | C1/C1S5 21           | P1-TBDI2-19    |
| XV-40 202A_DI_2 | C1/C1S5 11           | P1-TBDI2-11    |
| XV-40 202A_DO_1 | C1/C1S9 5            | P1-TBRL2-5     |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text              | Priority / Class    | Sounder Horn   |
|-------------------------------|---------------------|----------------|
| XV-40 202A SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| XV-40 202A WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| XV-40 202A ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
|---------------------|-----------------------|---------------------------|
| C1S5-21/P1-TBDI2-19 | P1-TBDI2-19/C1S5-21   | White (Sig), Grey (24VDC) |
| C1S5-11/P1-TBDI2-11 | P1-TBDI2-11/C1S5-11   | White (Sig), Grey (24VDC) |
| C1S9-5/P1-TBRL2-5   | P1-TBRL2-5/C1S9-5     | 24VDC Red (+), Blue (-)   |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Response Time | Result   | Remarks                       |
|---------------|----------|-------------------------------|
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |       |
|--|-------|
|  | KAL/  |
|  | Equip |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                      |
|----------------------|--------------------------------------|
| Equipment Tag:       | XV-40202B                            |
| Service Description: | Slurry feed-out pH Adjustment tank 2 |
| Instrument Type:     | Automatic on/off valve               |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                            | Core-to-Core         |
|----------------------------------------|------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test) | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\ \Omega$ end-to-end loop resistance  | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE      | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved           | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40202B | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-01                          |
| Calibrated Range:  | On-Off                                    |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 202B_DI_1 | C1/C1S5 12           | P1-TBDI2-12    |
| XV-40 202B_DI_2 | C1/C1S5 13           | P1-TBDI2-13    |
| XV-40 202B_DO_1 | C1/C1S9 6            | P1-TBRL2-6     |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text                 | Priority / Class    | Sounder Horn   |
|----------------------------------|---------------------|----------------|
| XV-40 202B SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| XV-40 202B WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| XV-40 202B ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S5-12/P1-TBDI2-12 | P1-TBDI2-12/C1S5-12   | White (Sig), Grey (24VDC) |
| C1S5-13/P1-TBDI2-13 | P1-TBDI2-13/C1S5-13   | White (Sig), Grey (24VDC) |
| C1S9-6/P1-TBRL2-6   | P1-TBRL2-6/C1S9-6     | 24VDC Red (+), Blue (-)   |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |       |
|--|-------|
|  | KAL/  |
|  | Equip |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                                |
|----------------------|------------------------------------------------|
| Equipment Tag:       | XV-40202C                                      |
| Service Description: | Water washing line outlet pH Adjustment tank 2 |
| Instrument Type:     | Automatic on/off valve                         |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                            | Core-to-Core         |
|----------------------------------------|------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test) | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\ \Omega$ end-to-end loop resistance  | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE      | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved           | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40202C | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-01                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 202C_DI_1 | C1/C1S5 14           | P1-TBDI2-14    |
| XV-40 202C_DI_2 | C1/C1S5 15           | P1-TBDI2-15    |
| XV-40 202C_DO_1 | C1/C1S9 7            | P1-TBRL2-7     |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text              | Priority / Class    | Sounder Horn   |
|-------------------------------|---------------------|----------------|
| XV-40 202C SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| XV-40 202C WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| XV-40 202C ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S5-14/P1-TBDI2-14 | P1-TBDI2-14/C1S5-14   | White (Sig), Grey (24VDC) |
| C1S5-15/P1-TBDI2-15 | P1-TBDI2-15/C1S5-15   | White (Sig), Grey (24VDC) |
| C1S9-7/P1-TBRL2-7   | P1-TBRL2-7/C1S9-7     | 24VDC Red (+), Blue (-)   |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |        |
|--|--------|
|  | KAL/   |
|  | Equip. |

### 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                         |
|----------------------|-----------------------------------------|
| Equipment Tag:       | XV-40202D                               |
| Service Description: | Slurry circulation pH Adjustment tank 2 |
| Instrument Type:     | Automatic on/off valve                  |

### 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

### 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

### 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

#### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

### 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40202D | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-01                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 202D_DI_1 | C1/C1S5 16           | P1-TBDI2-16    |
| XV-40 202D_DI_2 | C1/C1S5 19           | P1-TBDI2-17    |
| XV-40 202D_DO_1 | C1/C1S9 8            | P1-TBRL2-8     |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text              | Priority / Class    | Sounder Horn   |
|-------------------------------|---------------------|----------------|
| XV-40 202D SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| XV-40 202D WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| XV-40 202D ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S5-16/P1-TBDI2-16 | P1-TBDI2-16/C1S5-16   | White (Sig), Grey (24VDC) |
| C1S5-19/P1-TBDI2-17 | P1-TBDI2-17/C1S5-19   | White (Sig), Grey (24VDC) |
| C1S9-8/P1-TBRL2-8   | P1-TBRL2-8/C1S9-8     | 24VDC Red (+), Blue (-)   |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |       |
|--|-------|
|  | KAL   |
|  | Equip |

### 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                                              |
|----------------------|--------------------------------------------------------------|
| Equipment Tag:       | XV-40202E                                                    |
| Service Description: | Slurry pH Adjustment tank 2 transfer to pH Adjustment tank 1 |
| Instrument Type:     | Automatic on/off valve                                       |

### 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

### 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

### 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

#### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

### 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40202E | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-01                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 202E_DI_1 | C1/C1S5 20           | P1-TBDI2-18    |
| XV-40 202E_DI_2 | C1/C1S5 22           | P1-TBDI2-20    |
| XV-40 202E_DO_1 | C1/C1S8 19           | P1-TBRL1-17    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text              | Priority / Class    | Sounder Horn   |
|-------------------------------|---------------------|----------------|
| XV-40 202E SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| XV-40 202E WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| XV-40 202E ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S5-20/P1-TBDI2-18 | P1-TBDI2-18/C1S5-20   | White (Sig), Grey (24VDC) |
| C1S5-22/P1-TBDI2-20 | P1-TBDI2-20/C1S5-22   | White (Sig), Grey (24VDC) |
| C1S8-19/P1-TBRL1-17 | P1-TBRL1-17/C1S8-19   | 24VDC Red (+), Blue (-)   |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |       |
|--|-------|
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## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                               |
|----------------------|-------------------------------|
| Equipment Tag:       | XV-40202F                     |
| Service Description: | Slurry feed-to spray dry tank |
| Instrument Type:     | Automatic on/off valve        |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40202F | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-01                          |
| Calibrated Range:  | On-Off                                    |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 202F_DI_1 | C1/C1S5 23           | P1-TBDI2-21    |
| XV-40 202F_DI_2 | C1/C1S5 24           | P1-TBDI2-22    |
| XV-40 202F_DO_1 | C1/C1S8 20           | P1-TBRL1-18    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text              | Priority / Class    | Sounder Horn   |
|-------------------------------|---------------------|----------------|
| XV-40 202F SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| XV-40 202F WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| XV-40 202F ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

| CERTIFICATE |  |
|-------------|--|
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
|---------------------|-----------------------|---------------------------|
| C1S5-23/P1-TBDI2-21 | P1-TBDI2-21/C1S5-23   | White (Sig), Grey (24VDC) |
| C1S5-24/P1-TBDI2-22 | P1-TBDI2-22/C1S5-24   | White (Sig), Grey (24VDC) |
| C1S8-20/P1-TBRL1-18 | P1-TBRL1-18/C1S8-20   | 24VDC Red (+), Blue (-)   |

| Measured Value | Test Status | Inspector Initials |
|----------------|-------------|--------------------|
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

| Response Time | Result   | Remarks                       |
|---------------|----------|-------------------------------|
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

| Historian Log  | Result   | Remarks                 |
|----------------|----------|-------------------------|
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

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## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                        |
|----------------------|----------------------------------------|
| Equipment Tag:       | XV-40202G                              |
| Service Description: | CIP water washing pH Adjustment tank 2 |
| Instrument Type:     | Automatic on/off valve                 |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40202G | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-01                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 202G_DI_1 | C1/C1S5 25           | P1-TBDI2-23    |
| XV-40 202G_DI_2 | C1/C1S5 26           | P1-TBDI2-24    |
| XV-40 202G_DO_1 | C1/C1S8 21           | P1-TBRL1-19    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text              | Priority / Class    | Sounder Horn   |
|-------------------------------|---------------------|----------------|
| XV-40 202G SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| XV-40 202G WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| XV-40 202G ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

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| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S5-25/P1-TBDI2-23 | P1-TBDI2-23/C1S5-25   | White (Sig), Grey (24VDC) |
| C1S5-26/P1-TBDI2-24 | P1-TBDI2-24/C1S5-26   | White (Sig), Grey (24VDC) |
| C1S8-21/P1-TBRL1-19 | P1-TBRL1-19/C1S8-21   | 24VDC Red (+), Blue (-)   |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

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|  | Equip. |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                                       |
|----------------------|-------------------------------------------------------|
| Equipment Tag:       | XV-40202H                                             |
| Service Description: | CIP Hight pressure water washing pH Adjustment tank 2 |
| Instrument Type:     | Automatic on/off valve                                |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40202H | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-01                          |
| Calibrated Range:  | On-Off                                    |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 202H_DI_1 | C1/C1S5 27           | P1-TBDI2-25    |
| XV-40 202H_DI_2 | C1/C1S5 28           | P1-TBDI2-26    |
| XV-40 202H_DO_1 | C1/C1S8 22           | P1-TBRL1-20    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text              | Priority / Class    | Sounder Horn   |
|-------------------------------|---------------------|----------------|
| XV-40 202H SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| XV-40 202H WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| XV-40 202H ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

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| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S5-27/P1-TBDI2-25 | P1-TBDI2-25/C1S5-27   | White (Sig), Grey (24VDC) |
| C1S5-28/P1-TBDI2-26 | P1-TBDI2-26/C1S5-28   | White (Sig), Grey (24VDC) |
| C1S8-22/P1-TBRL1-20 | P1-TBRL1-20/C1S8-22   | 24VDC Red (+), Blue (-)   |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

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|  | Equip |

### 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                         |
|----------------------|-----------------------------------------|
| Equipment Tag:       | XV-40203B                               |
| Service Description: | Slurry feed-out Spray dryer infeed tank |
| Instrument Type:     | Automatic on/off valve                  |

### 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

### 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

### 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

#### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

### 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40203B | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-02                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 203B_DI_1 | C1/C1S5 29           | P1-TBDI2-27    |
| XV-40 203B_DI_2 | C1/C1S5 30           | P1-TBDI2-28    |
| XV-40 203B_DO_1 | C1/C1S8 23           | P1-TBRL1-21    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text                 | Priority / Class    | Sounder Horn   |
|----------------------------------|---------------------|----------------|
| XV-40 203B SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| XV-40 203B WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| XV-40 203B ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S5-29/P1-TBDI2-27 | P1-TBDI2-27/C1S5-29   | White (Sig), Grey (24VDC) |
| C1S5-30/P1-TBDI2-28 | P1-TBDI2-28/C1S5-30   | White (Sig), Grey (24VDC) |
| C1S8-23/P1-TBRL1-21 | P1-TBRL1-21/C1S8-23   | 24VDC Red (+), Blue (-)   |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |       |
|--|-------|
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|  | Equip |

### 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                                   |
|----------------------|---------------------------------------------------|
| Equipment Tag:       | XV-40203C                                         |
| Service Description: | Water washing line outlet Spray dryer infeed tank |
| Instrument Type:     | Automatic on/off valve                            |

### 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

### 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

### 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

#### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

### 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40203C | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |  |                                           |
|--------------------|--|-------------------------------------------|
| P&ID Number:       |  | 325-01-402-PD-02                          |
| Calibrated Range:  |  | On-Off                                    |
| Branch Cable Spec: |  | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 203C_DI_1 | C1/C1S5 31           | P1-TBDI2-29    |
| XV-40 203C_DI_2 | C1/C1S5 32           | P1-TBDI2-30    |
| XV-40 203C_DO_1 | C1/C1S8 24           | P1-TBRL1-22    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text                 | Priority / Class    | Sounder Horn   |
|----------------------------------|---------------------|----------------|
| XV-40 203C SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| XV-40 203C WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| XV-40 203C ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S5-31/P1-TBDI2-29 | P1-TBDI2-29/C1S5-31   | White (Sig), Grey (24VDC) |
| C1S5-32/P1-TBDI2-30 | P1-TBDI2-30/C1S5-32   | White (Sig), Grey (24VDC) |
| C1S8-24/P1-TBRL1-22 | P1-TBRL1-22/C1S8-24   | 24VDC Red (+), Blue (-)   |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |        |
|--|--------|
|  | KAL/   |
|  | Equip. |

## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                           |
|----------------------|-------------------------------------------|
| Equipment Tag:       | XV-40203D                                 |
| Service Description: | CIP water washing Spray dryer infeed tank |
| Instrument Type:     | Automatic on/off valve                    |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                 | Core-to-Core         |
|----------------------------------------|-----------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE           | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40203D | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-02                          |
| Calibrated Range:  | On-Off                                    |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 203D_DI_1 | C1/C1S5 33           | P1-TBDI2-31    |
| XV-40 203D_DI_2 | C1/C1S5 34           | P1-TBDI2-32    |
| XV-40 203D_DO_1 | C1/C1S8 25           | P1-TBRL1-23    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text                 | Priority / Class    | Sounder Horn   |
|----------------------------------|---------------------|----------------|
| XV-40 203D SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| XV-40 203D WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| XV-40 203D ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S5-33/P1-TBDI2-31 | P1-TBDI2-31/C1S5-33   | White (Sig), Grey (24VDC) |
| C1S5-34/P1-TBDI2-32 | P1-TBDI2-32/C1S5-34   | White (Sig), Grey (24VDC) |
| C1S8-25/P1-TBRL1-23 | P1-TBRL1-23/C1S8-25   | 24VDC Red (+), Blue (-)   |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |       |
|--|-------|
|  | KAL   |
|  | Equip |

### 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                                          |
|----------------------|----------------------------------------------------------|
| Equipment Tag:       | XV-40203E                                                |
| Service Description: | CIP hight pressure water washing Spray dryer infeed tank |
| Instrument Type:     | Automatic on/off valve                                   |

### 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                | Type |
|-------|-----------------------------------------------|------|
| CH-01 | Limit Switch - Full Open Feedback (ZSO / DI)  | DI   |
| CH-02 | Limit Switch - Full Close Feedback (ZSC / DI) | DI   |
| CH-03 | Solenoid Valve Actuation Command (24VDC DO)   | DO   |

### 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                            | Core-to-Core         |
|----------------------------------------|------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20\text{ M}\Omega$ @ 500VDC (1 min test) | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0\ \Omega$ end-to-end loop resistance  | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE      | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved           | Engraved SS          |

### 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

#### 4.1 DISCRETE ACTUATION & FEEDBACK STATUS VERIFICATION

| Signal Item           | Command / State      | Field Voltage / Contact           |
|-----------------------|----------------------|-----------------------------------|
| Solenoid Command (DO) | Energize to Open     | 24.0 VDC @ Coil                   |
| Open Feedback (ZSO)   | Valve Fully Open     | Dry Contact Closed ( $<2\Omega$ ) |
| Solenoid Command (DO) | De-energize to Close | 0.0 VDC @ Coil                    |
| Close Feedback (ZSC)  | Valve Fully Closed   | Dry Contact Closed ( $<2\Omega$ ) |

### 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Chime / Fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: XV-40203E | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                           |
|--------------------|-------------------------------------------|
|                    |                                           |
| P&ID Number:       | 325-01-402-PD-02                          |
| Calibrated Range:  | On-Off                                    |
| Branch Cable Spec: | 1Px0.75mm² Shielded Instrumentation Cable |

| PLC Tag Name    | PLC Hardware Address | JB Terminal ID |
|-----------------|----------------------|----------------|
| XV-40 203E_DI_1 | C1/C1S6 1            | P1-TBDI3-1     |
| XV-40 203E_DI_2 | C1/C1S6 2            | P1-TBDI3-2     |
| XV-40 203E_DO_1 | C1/C1S8 26           | P1-TBRL1-24    |

| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
|----------------|---------------|-----------------|
| ____ MΩ        | ____ MΩ       | ____ MΩ         |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

| PLC Bit Status         | SCADA Graphic State   | Feedback Travel Angle |
|------------------------|-----------------------|-----------------------|
| Bit = 1 (Active)       | Graphic Green (Open)  | 90° Full Travel       |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Open Target Hit       |
| Bit = 0 (De-energized) | Graphic Red (Closed)  | 0° Full Travel        |
| Bit = 1 (Active)       | Limit Switch Icon Lit | Close Target Hit      |

| SCADA Alarm Text              | Priority / Class    | Sounder Horn   |
|-------------------------------|---------------------|----------------|
| XV-40 203E SAFETY SHUTDOWN    | High (Priority 1)   | Audible Active |
| XV-40 203E WIRE BREAK / FAULT | Medium (Priority 2) | Warning Tone   |
| XV-40 203E ALARM ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
| TIFICATE    |  |
| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 3 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                     |                       |                           |
|---------------------|-----------------------|---------------------------|
|                     |                       |                           |
| Wire Tag (PLC Side) | Wire Tag (Field Side) | Core Wiring Colors        |
| C1S6-1/P1-TBDI3-1   | P1-TBDI3-1/C1S6-1     | White (Sig), Grey (24VDC) |
| C1S6-2/P1-TBDI3-2   | P1-TBDI3-2/C1S6-2     | White (Sig), Grey (24VDC) |
| C1S8-26/P1-TBRL1-24 | P1-TBRL1-24/C1S8-26   | 24VDC Red (+), Blue (-)   |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|               |          |                               |
|---------------|----------|-------------------------------|
|               |          |                               |
|               |          |                               |
| Response Time | Result   | Remarks                       |
| < 1.2 sec     | [ ] PASS | Pneumatic travel smooth       |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |
| < 1.5 sec     | [ ] PASS | Spring return action verified |
| < 0.8 sec     | [ ] PASS | Proximity target aligned      |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

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## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                     |
|----------------------|-------------------------------------|
| Equipment Tag:       | pHT-40201                           |
| Service Description: | pH analyzer in pH Adjustment tank 1 |
| Instrument Type:     | pH Analyzer                         |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                                        | Type |
|-------|-----------------------------------------------------------------------|------|
| CH-01 | Analog Input Process Variable (4-20mA AI) - PHT-40201_AI_1 [Pins A+B] | AI   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                  | Core-to-Core         |
|----------------------------------------|------------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20 \text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0 \text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE            | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                 | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 ANALOG 5-POINT CALIBRATION TABLE (Current Loop Simulation: Pin A = IN-x, Pin B = i RTN-)

| Calibration Point | Simulated Input (mA) | Field Indicator Value |
|-------------------|----------------------|-----------------------|
| 0% (Zero)         | 4.000 mA             | 4.00 mA               |
| 25% (Span 1/4)    | 8.000 mA             | 8.01 mA               |
| 50% (Mid)         | 12.000 mA            | 12.00 mA              |
| 75% (Span 3/4)    | 16.000 mA            | 15.99 mA              |
| 100% (Full Span)  | 20.000 mA            | 20.00 mA              |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                     |
|---------------------------------|--------------------------------------------|------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state    |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Channel fault banner / Hold permissive   |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ASIN ENGINEERING | INSTRUMENT LOOP TEST & CALIBRATION CERTIFICATE

Instrument: pH<sub>T</sub>-40201 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

|                    |                                                       |
|--------------------|-------------------------------------------------------|
|                    |                                                       |
| P&ID Number:       | 325-01-402-PD-01                                      |
| Calibrated Range:  | 0-14                                                  |
| Branch Cable Spec: | 1Px0.75mm <sup>2</sup> Shielded Instrumentation Cable |

|                 |                             |                 |
|-----------------|-----------------------------|-----------------|
|                 |                             |                 |
| PLC Tag Name    | PLC Hardware Address        | JB Terminal ID  |
| pHT-40 201_AI_1 | C1/C1S10 Pins 16(A) & 15(B) | P1-TBAI1-7(A/B) |

|                |               |                 |
|----------------|---------------|-----------------|
|                |               |                 |
| Core-to-Shield | Core-to-Earth | Shield-to-Earth |
| _____ MΩ       | _____ MΩ      | _____ MΩ        |
| -              | -             | -               |
| -              | -             | Isolated        |
| Gland Tight    | Drain Plug OK | O-ring Intact   |

|                |                    |                    |
|----------------|--------------------|--------------------|
|                |                    |                    |
| c)             |                    |                    |
| PLC Raw Counts | SCADA Indicated EU | Calculated Error % |
| 0 counts       | 0.0 EU             | +0.00%             |
| 8192 counts    | 25.0 EU            | +0.05%             |
| 16 384 counts  | 50.0 EU            | +0.00%             |
| 24 576 counts  | 74.9 EU            | -0.05%             |
| 32 767 counts  | 100.0 EU           | +0.00%             |

|                                  |                     |                |
|----------------------------------|---------------------|----------------|
|                                  |                     |                |
| SCADA Alarm Text                 | Priority / Class    | Sounder Horn   |
| pHT-40 201 SAFETY SHUTDOWN       | High (Priority 1)   | Audible Active |
| pHT-40 201 WIRE BREAK / FAULT    | Medium (Priority 2) | Warning Tone   |
| pHT-40 201 ALARM<br>ACKNOWLEDGED | Low (Priority 3)    | Silenced       |

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| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

|             |  |
|-------------|--|
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| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                                                         |                                                         |                                       |
|---------------------------------------------------------|---------------------------------------------------------|---------------------------------------|
|                                                         |                                                         |                                       |
| Wire Tag (PLC Side)                                     | Wire Tag (Field Side)                                   | Core Wiring Colors                    |
| C1S10-16/P1-TBAI1-7(A) (+) / C1S10-15/P1-TBAI1-7(B) (-) | P1-TBAI1-7(A)/C1S10-16 (+) / P1-TBAI1-7(B)/C1S10-15 (-) | White (+ / IN-x), Black (- / i RTN-x) |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|                     |          |                                 |
|---------------------|----------|---------------------------------|
|                     |          |                                 |
|                     |          |                                 |
| Max Allowable Error | Result   | Remarks                         |
| ± 0.20%             | [ ] PASS | Fluke 789 Injected Across A & B |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Span limit verified             |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

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| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |

|  |       |
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## 1. EQUIPMENT IDENTIFICATION & PROCESS METADATA

|                      |                                     |
|----------------------|-------------------------------------|
| Equipment Tag:       | pHT-40202                           |
| Service Description: | pH analyzer in pH Adjustment tank 2 |
| Instrument Type:     | pH Analyzer                         |

## 2. FIELD WIRING & PLC I/O CHANNEL ALLOCATION (AI: 2-PIN DIFFERENTIAL PAIR)

| CH #  | Channel Function & Description                                        | Type |
|-------|-----------------------------------------------------------------------|------|
| CH-01 | Analog Input Process Variable (4-20mA AI) - PHT-40202_AI_1 [Pins A+B] | AI   |

## 3. PHYSICAL INSPECTION & COLD LOOP INTEGRITY VERIFICATION

| Cold Test Item                         | Acceptance Criteria                                  | Core-to-Core         |
|----------------------------------------|------------------------------------------------------|----------------------|
| Cable Insulation Resistance (Megger)   | $\geq 20 \text{ M}\Omega$ @ 500VDC (1 min test)      | ___ $\text{M}\Omega$ |
| Loop Conductor Resistance (Continuity) | $\leq 2.0 \text{ }\Omega$ end-to-end loop resistance | ___ $\Omega$         |
| Single-Point Earth Shielding           | Isolated at Field; Single-point at CA1 IE            | -                    |
| Enclosure, Gland & Tag Inspection      | IP66/67 Gland tight, SS Tag Engraved                 | Engraved SS          |

## 4. LIVE SIGNAL CALIBRATION & FUNCTIONAL ACTUATION VERIFICATION

### 4.1 ANALOG 5-POINT CALIBRATION TABLE (Current Loop Simulation: Pin A = IN-x, Pin B = i RTN-)

| Calibration Point | Simulated Input (mA) | Field Indicator Value |
|-------------------|----------------------|-----------------------|
| 0% (Zero)         | 4.000 mA             | 4.00 mA               |
| 25% (Span 1/4)    | 8.000 mA             | 8.01 mA               |
| 50% (Mid)         | 12.000 mA            | 12.00 mA              |
| 75% (Span 3/4)    | 16.000 mA            | 15.99 mA              |
| 100% (Full Span)  | 20.000 mA            | 20.00 mA              |

## 5. SAFETY INTERLOCK, EMERGENCY TRIP & SCADA ALARM RESPONSE

| Safety Test Item                | Trigger Condition                          | Expected Safe Action                                |
|---------------------------------|--------------------------------------------|-----------------------------------------------------|
| Emergency Stop / Trip Interlock | E-Stop Pushbutton or SIS Trip              | Fail-safe spring return to safe state               |
| Out-of-Range Alarm              | Signal $<3.6\text{mA}$ or $>21.0\text{mA}$ | Channel fault banner / Hold                         |
| SCADA Alarm Acknowledge         | Operator Acknowledge click                 | permissive Banner silences horn, latches steady red |

## 6. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

| Sign-off Role                          | Name & Designation             | Company / Organization         |
|----------------------------------------|--------------------------------|--------------------------------|
| <b>Tested by (Contractor):</b>         | Lead Instrument Engineer       | Kalasin Engineering Co., Ltd.  |
| <b>Witnessed by (Client QC):</b>       | Senior Electrical & Inst QA/QC | Ingredion (Thailand) Co., Ltd. |
| <b>Approved by (Commissioning PM):</b> | Project Commissioning Manager  | Kalasin / Ingredion Joint Team |

ment: pHT-40202 | Junction Box: JB-401 | Ref Std: IEC 62381 / ISA-RP60.8 (AI: 2-Pin Differential)

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| Signature | Date Signed | Overall Verdict     |
|-----------|-------------|---------------------|
| _____     | DD-MM-2026  | LOOP CHECK PASSED   |
| _____     | DD-MM-2026  | ACCEPTED & APPROVED |
| _____     | DD-MM-2026  | FINAL AUTHORIZED    |

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|-------------|--|
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| al Channel) |  |

|                  |                              |
|------------------|------------------------------|
|                  |                              |
| Junction Box:    | JB-401                       |
| Channel Count:   | 1 Active Process Channels    |
| CAD Drawing Ref: | LOOP DIAGRAM WRING.dwg / dxf |

|                                                         |                                                         |                                       |
|---------------------------------------------------------|---------------------------------------------------------|---------------------------------------|
|                                                         |                                                         |                                       |
| Wire Tag (PLC Side)                                     | Wire Tag (Field Side)                                   | Core Wiring Colors                    |
| C1S10-18/P1-TBAI1-8(A) (+) / C1S10-17/P1-TBAI1-8(B) (-) | P1-TBAI1-8(A)/C1S10-18 (+) / P1-TBAI1-8(B)/C1S10-17 (-) | White (+ / IN-x), Black (- / i RTN-x) |

|                |             |                    |
|----------------|-------------|--------------------|
|                |             |                    |
| Measured Value | Test Status | Inspector Initials |
| > 100 MΩ       | [ ] PASS    | -----              |
| 0.45 Ω         | [ ] PASS    | -----              |
| CUTBACK & TAPE | [ ] PASS    | -----              |
| Inspected      | [ ] PASS    | -----              |

|                     |          |                                 |
|---------------------|----------|---------------------------------|
|                     |          |                                 |
|                     |          |                                 |
| Max Allowable Error | Result   | Remarks                         |
| ± 0.20%             | [ ] PASS | Fluke 789 Injected Across A & B |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Linearity verified              |
| ± 0.20%             | [ ] PASS | Span limit verified             |

|                |          |                         |
|----------------|----------|-------------------------|
|                |          |                         |
| Historian Log  | Result   | Remarks                 |
| Logged with ms | [ ] PASS | Hardwired trip verified |
| Logged with ms | [ ] PASS | NAMUR NE43 compliance   |
| Logged with ms | [ ] PASS | HMI banner functioning  |

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

| Punchlist Ref | Tag Color Affixed  | Remarks                           |
|---------------|--------------------|-----------------------------------|
| None (Clean)  | Green Loop Tag     | Ready for Process Commissioning   |
| None          | QC Sticker Applied | Satisfies Contract Scope          |
| None          | System Cleared     | Authorized for Cold Water Wet Run |